UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF INFORMATION SYSTEMS

FINAL PROJECT REPORT
DATA VISUALIZATION

GROUP: 7

From Data to Decisions: A Full BI and Visualization Approach to Loan Analysis

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Finally, we would like to thank our classmates for their collaboration, discussions, and motivation. This report is the result of the collective effort and commitment of Group 7, and we truly appreciate everyone who contributed to its success.

Group 7

Commitment

We, Group 7, hereby declare that our study titled "Data Visualization and Analytical Insights for Loan Default Analysis" is the result of our own independent work, carried out under the scientific supervision of MSc. Lê Bá Thiên.

All data, analyses, and findings presented in this report have been collected and processed with integrity, accuracy, and transparency. References to external sources, including both domestic and international studies, have been properly cited and listed in the reference section.

While we have made every effort to ensure the quality and validity of our research, certain limitations may still exist due to time and data constraints. We sincerely welcome any feedback or suggestions to improve and strengthen this study.

Group 7

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List of Abbreviations

Abbreviation	Description
BI	Business Intelligence
DW	Data Warehouse
DV	Data Visualization
ML	Machine Learning
ETL	Extract, Transform, Load
SQL	Structured Query Language
CRM	Customer Relationship Management
DBMS	Database Management System
KPI	Key Performance Indicator
YOY	Year Over Year
DAX	Data Analysis Expressions
SMOTE	Synthetic Minority Oversampling Technique
EDA	Exploratory Data Analysis
DTIRatio	Debt-to-Income Ratio
NPL	Non-Performing Loan
APR	Annual Percentage Rate
AI	Artificial Intelligence
SHAP	SHapley Additive exPlanations
GBM	Gradient Boosting Machine
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine

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CHAPTER 1 Introduction

1.1 Business case of the projects

The modern financial sector requires robust credit risk management to maintain stability and profitability. However, many lending institutions still rely on manual or static methods for loan applicant evaluation. This project addresses several key business problems that arise from these outdated practices:

- Inefficient Risk Assessment: Current credit evaluation systems frequently use static criteria or manual scoring, failing to capture complex, non-linear risk patterns and behavioral factors that significantly influence a borrower's likelihood to default. This methodology prevents the accurate forecasting of potential defaults.
- Fragmented and Non-Standardized Data: Loan and borrower information is often scattered across various systems (CRM, accounting, risk databases). This fragmentation hinders the establishment of a unified, 360-degree view of the customer, and inconsistencies in data types and formats complicate the integration necessary for effective analytical modeling.
- Lack of Predictive Intelligence (Reactive Reporting): Existing Business Intelligence (BI) tools primarily deliver historical summaries (descriptive analytics), showing what has happened, such as past default rates. This reactive approach limits the ability to implement preventive strategies and detect potential risks early.
- Strategic Need for Data-Driven Decision-Making: To remain competitive, financial institutions must shift from descriptive reporting to predictive and prescriptive analytics.

Therefore, the business case for this project is to develop an integrated analytical ecosystem powered by a Data Warehouse (DW) foundation, Data Visualization (DV) techniques, and Machine Learning (ML) models. The project leverages historical loan data to generate actionable insights that improve the accuracy and transparency of credit risk assessment.

Data Visualization plays a central role throughout the analytical workflow. It is applied not only to explore and evaluate data extracted from the DW but also to enhance feature selection, model training, and performance evaluation. By combining predictive modeling (e.g., Logistic Regression, Random Forest, Gradient Boosting) with interactive visualization, the system enables analysts to interpret model behavior intuitively, identify key risk factors, and refine model parameters more effectively.

This integration of DW, DV, and ML transforms traditional Business Intelligence from a descriptive tool into a proactive decision-support framework. It empowers financial institutions to detect potential defaulters early, optimize lending strategies, and foster data-driven, insight-oriented risk management.

1.2 Objectives of the Project

1.2.1 General Objective

The general objective of this project is to develop an integrated Business Intelligence (BI) system that combines Data Warehousing (DW), Data Visualization (DV), and Machine Learning (ML) to improve credit risk assessment and loan default prediction. Through this system, financial institutions can transform historical and fragmented loan data into meaningful insights that enhance risk monitoring, optimize lending strategies, and support data-driven decision-making.

1.2.2 Specific Objectives

To achieve the overall goal, this project aims to implement a centralized Data Warehouse following the Star Schema design to ensure efficient data integration and consistency across different borrower and loan attributes. The project also applies the ETL (Extract, Transform, Load) process to clean, transform, and standardize the dataset for analytical use.

In addition, the project conducts exploratory and descriptive analyses using BI tools such as Power BI and Python to identify key factors influencing default risk. Predictive modeling techniques, including Logistic Regression, Random Forest, and Gradient Boosting, are employed to estimate the likelihood of loan default based on both financial and demographic variables.

Furthermore, the project integrates predictive results into interactive dashboards to visualize trends, risk patterns, and borrower profiles in an intuitive way. By combining analytics and visualization, the project ultimately seeks to provide a reliable decision-support system that helps credit managers detect high-risk cases early and improve lending performance.

1.2.3 Business Questions

To guide the analysis and ensure practical outcomes, the project focuses on addressing the following key business questions:

- How have total loan amounts and default rates changed over time?
- Which borrower characteristics and financial indicators are most related to default risk?
- What demographic or employment groups show higher default tendencies? What is the predicted probability of default for each borrower segment?

● How can BI and ML insights support better lending and credit management decisions?

1.3 Research Object

The research object of this Business Intelligence (BI) project focuses on the Loan Default Dataset, which contains information about personal loans and borrowers' repayment behavior. The dataset includes detailed records for 47 loan profiles (based on the provided data image), with key attributes such as: age (Age), annual income (Income), loan amount (LoanAmount), credit score (CreditScore), months employed (MonthsEmployed), number of active credit lines (NumCreditLines), interest rate (InterestRate), loan term (LoanTerm), debt-to-income ratio (DTIRatio), education level (Education), employment type (EmploymentType), marital status (MaritalStatus), mortgage status (HasMortgage), dependents status (HasDependents), loan purpose (LoanPurpose), co-signer status (HasCoSigner), and default outcome (Default). The loan dates (Loan Date) are recorded from 2013 to 2018, in the format DD/MM/YYYY.

The objective of the research is applying analytics to explore factors influencing default risk, such as the relationship between low credit scores, high DTIRatio, or unstable employment types and the likelihood of default. Through BI analysis, the project will develop predictive models, visual dashboards, and insightful reports to assist financial institutions (e.g., banks) in evaluating credit risk, optimizing loan approval processes, and minimizing losses from defaults.

1.4 Scope of the Project

The scope of this project is limited to the exploitation and analysis of the provided Loan Default Dataset, focusing on the following activities:

● Data Analysis: Utilizing BI tools (such as Power BI, Tableau, or Python with libraries like pandas and matplotlib) to perform descriptive analytics, diagnostic analytics, and predictive analytics on default risks. This includes identifying strong correlations with the Default variable, such as CreditScore below 500 or DTIRatio above 0.4.

● Model Building: Developing basic machine learning models (e.g., logistic regression or decision trees) to predict default probability based on input variables like Age, Income, and LoanPurpose.

● Visualization and Reporting: Creating dashboards with charts (e.g., default rates by Education level or EmploymentType), summary tables, and insight reports for end-users (e.g., credit risk managers).

● Data Timeline: Using only historical data from 2013 to 2018, without incorporating real-time or updated data.

1.5 Expected Value and Outcomes:

This project, grounded in the analysis of the Loan Default Dataset (data covering 2013 to 2018), is expected to deliver substantial value by enhancing credit risk management capabilities.

The primary expected outcomes are grouped into analytical, technical, and strategic values:

Table 1. 1. Expect Value Objects

Category	**Expected Value and Outcome**
I. Analytical & Diagnostic Value	Insightful Reports and Descriptive Analytics: The project will produce insightful reports and summary statistics by performing descriptive and diagnostic analytics on Default risks, Loan Amount, Interested Rate. This includes creating dashboards that show default rates, Loan Amount, Interested Rate segmented by key categorical dimensions like Education level, Employment Type, Credit Score.
	Factor Identification: The analysis aims to explore and identify key factors influencing default risk, such as the relationship between low credit scores, high DTIRatio and the likelihood of default.

II. Predictive Value	Predictive Modeling Capability: By developing machine learning models (such as Logistic Regression or Decision Trees), the project will provide the ability to predict the probability of default based on input variables like Age, Income, and Loan Purpose.
	Enhanced Risk Evaluation: The predictive models will allow financial institutions to detect potential defaulters earlier, enhancing credit-risk evaluation and making lending decisions more data-driven.
III. Technical & Strategic Value	Centralized Data Repository: Successful implementation of the Data Warehouse (DW) and Star Schema ensures that loan and borrower data are consolidated and transformed into a format optimized for reporting and analysis, enhancing query performance for end-users.
	Optimized Decision-Making: Through visual dashboards, managers can understand business performance quickly, allowing for faster problem detection and opportunity spotting, ultimately optimizing the loan approval processes and minimizing losses from defaults.

1.6 Structure of Report

This report consists of seven chapters, each describing a key stage in the development of the Business Intelligence (BI) system for credit risk analysis.

Chapter 1 – Introduction outlines the background, business case, project objectives, research scope, and expected outcomes.

Chapter 2 – Theoretical Basis provides an overview of the theoretical concepts related to Data Visualization, Analytics, ETL processes, Data Warehousing, and Machine Learning that form the foundation of the project.

Chapter 3 – Requirement Analysis defines the business problems, data sources, data challenges, and key performance indicators (KPIs) for evaluating credit risk.

Chapter 4 – Data Preparation and Data Warehouse Implementation describes the data cleaning, transformation, and integration process, as well as the design of the Data Warehouse using the Star Schema model.

Chapter 5 – Data Visualization in Power BI presents the creation of interactive dashboards and analytical visualizations that illustrate loan performance, borrower characteristics, and default trends.

Chapter 6 – Machine Learning Model and Insight Generation explains the development and evaluation of predictive models, comparing algorithms such as Logistic Regression, Random Forest, and Gradient Boosting, and integrating results into BI dashboards.

Chapter 7 – Conclusion, Limitations, and Future Recommendations summarizes the main findings, discusses project limitations, and proposes future directions such as data expansion, advanced model enhancement, and real-time predictive analytics.

CHAPTER 2 Theoretical Basis

2.1 Data Visualization and Analytics

2.1.1 Data Visualization

Data visualization is a powerful tool for exploring data to more easily identify patterns, recognize anomalies or irregularities in the data, and better understand the relationships between variables. Our ability to spot these types of characteristics of data is much stronger and quicker when we look at a visual display of the data rather than a simple listing.

2.1.2 Analytics

Analytics is the scientific process of transforming data into insights for making better decisions. Analytics is generally grouped into three broad categories of methods: descriptive, predictive, and prescriptive analytics:

- Descriptive analytics is the set of analytical tools that describe what has happened. This includes techniques such as data queries (requests for information with certain characteristics from a database), reports, descriptive or summary statistics, and data visualization.
- Predictive analytics consists of techniques that use mathematical models constructed from past data to predict future events or better understand the relationships between variables.
- Prescriptive analytics are mathematical or logical models that suggest a decision or course of action. This category includes mathematical optimization models, decision analysis, and heuristic or rule-based systems

2.2 ETL Process

2.2.1 What is ETL?

ETL (Extract, Transform, Load) is crucial to the process of transforming and normalizing data from various sources into a format that is suitable for storing and analyzing for purposes of data management or analysis. This procedure involves three main steps: take the extract, transform, and load.

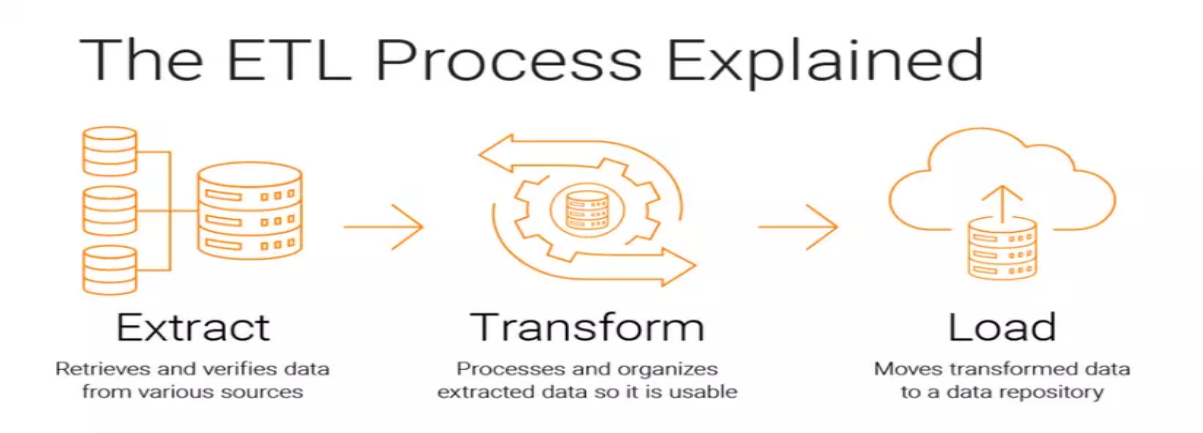

Figure 2. 1. ETL extract transform load

2.2.2 ETL Process

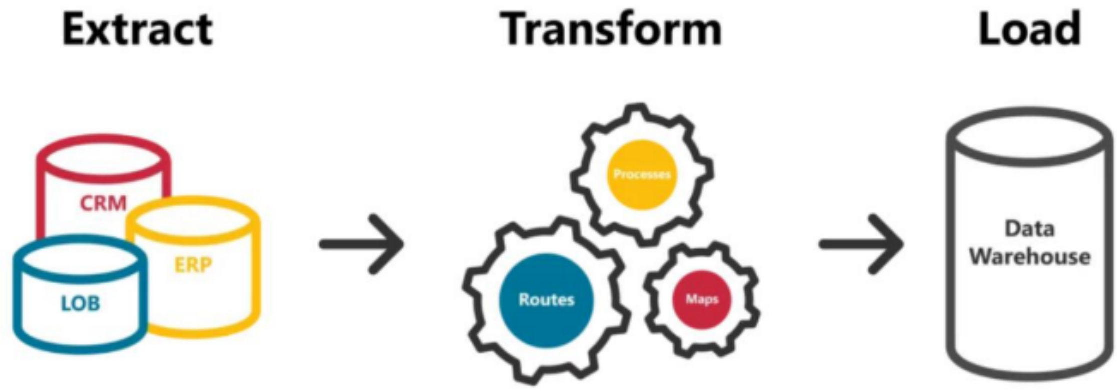

Figure 2. 2. ETL Process using Data Warehouse

The ETL process is following these steps:

- Extract: This procedure involves accessing different data sources in order to gather the data and store it temporarily. For database systems, the extraction process can be accomplished using SQL or another database management software to access data from tables or other data structures. For data that is stored in files, ETL can utilize file reading tools to read data from CSV, Excel, or other data storage types.
- Transform: This procedure involves converting the extracted information into a form that is suitable for the intended goal. This procedure involves operations like deleting duplicate data, replacing null or missing values, converting data types, creating calculated columns, and so on.
- Load: This step is the process of transferring the processed information into the main storage location in order to achieve the desired goal. The repository can be a relational database or other storage systems like a data lake or data warehouse.

2.3 Data Warehouse

A data warehouse (DW) is a system that consolidates data from various sources and environments, such as sales software, accounting, human resources, or core banking systems. It enhances query performance for reporting and analysis, grouping data to support decision-making. Serving as a central repository, a data warehouse collects data from transaction systems and related databases. This data is then processed and transformed, enabling users to access it through Business Intelligence tools, SQL clients, or spreadsheets.

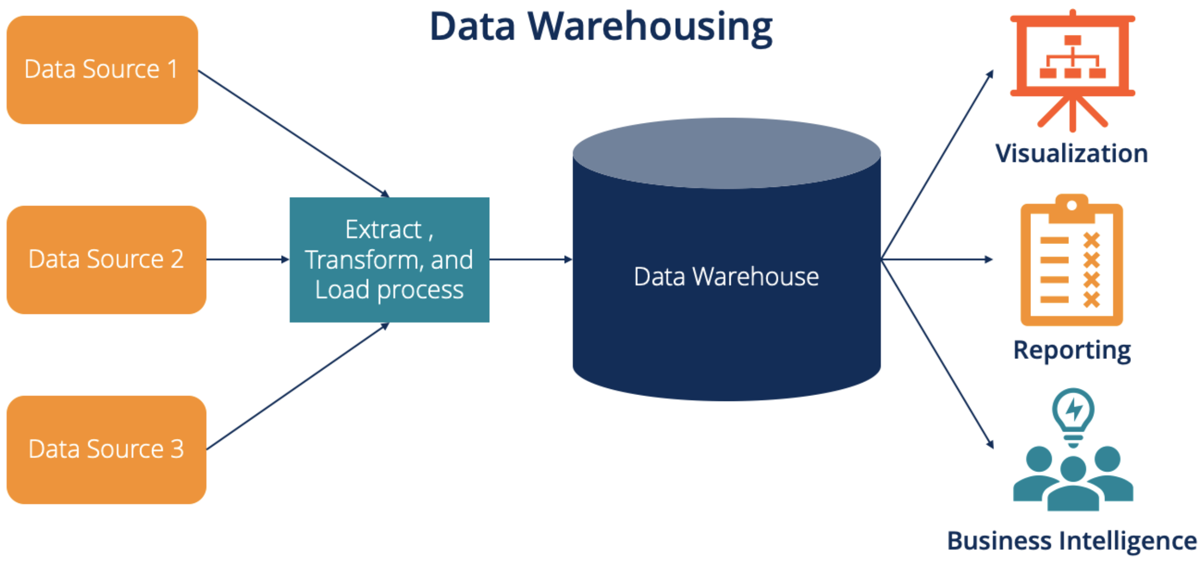

Figure 2. 3. Data Warehouse

Star Schema:

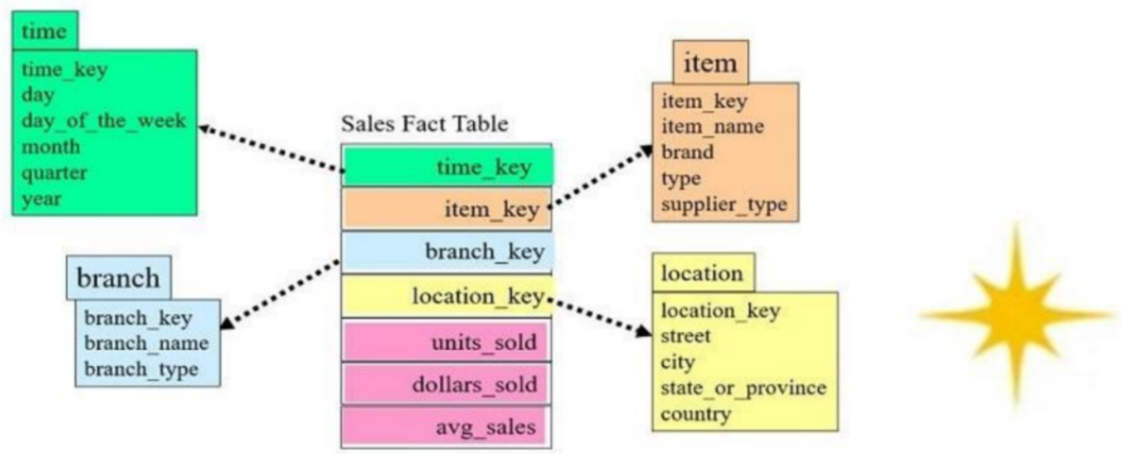

Figure 2. 4. Star schema

In the example shown in Figure 2.4, the event table contains sales data with metrics such as units sold, dollars sold, and average sales. This event table is linked to four dimension tables: time, store branch, item, and store location. A key feature of the star model is that each dimension table is directly related to the fact table, which allows the

DBMS to process and return data quickly. However, the downside of this approach is that some dimension tables are not normalized.

The project adopts a Star Schema approach to design and implement a Data Warehouse (DW) based on the Loan Default Dataset. This data modeling technique is chosen to optimize query performance and provide analytical flexibility.

The Star Schema consists of a central fact table (DWLoan.FactLoan) that stores key quantitative metrics (e.g., CreditScore, DTIRatio, LoanAmount), directly linked to several dimension tables through surrogate keys. This structure allows the Database Management System (DBMS) to efficiently process and retrieve data—an essential requirement for conducting descriptive and diagnostic analyses within Business Intelligence (BI) tools.

From a data processing perspective, this approach follows the Dimensional Modeling methodology, which transforms raw, fragmented, and non-normalized data into a structured analytical model. It separates descriptive attributes (dimensions) from quantitative measures (facts), ensuring both clarity and analytical consistency.

2.4 Data Visualization Concepts

2.4.1 Importance of Visualization in Business with Data

Data visualization plays a very important role in business today. In a world full of data, it helps people see patterns, trends, and relationships that are not obvious from tables or numbers alone. Instead of reading long reports, managers and decision makers can understand business performance quickly by looking at charts, dashboards, and maps.

Visualization helps businesses in many ways:

- Better decision-making: When data is visualized clearly, decision makers can detect problems and opportunities faster. For example, a sales dashboard may show which products perform best in each region.
- Detecting trends: Line charts or bar charts can reveal seasonal trends or customer behavior patterns.
- Performance monitoring: Dashboards allow tracking of key performance indicators (KPIs) in real time, helping companies take quick actions.
- Communication and storytelling: Visuals make it easier to share insights with non-technical audiences. A clear chart can explain a business situation better than a spreadsheet full of numbers.

In short, visualization turns complex data into an easy-to-understand visual story that supports faster and smarter business actions.

2.4.2 Best Practices in Power BI Dashboard

A Power BI dashboard is a visual interface that combines charts, KPIs, and reports to provide a quick overview of business data. To create effective dashboards, following best practices in design and visualization is essential.

1. Keep it simple and focused

Avoid adding too many visuals. Focus only on the most important KPIs or questions the dashboard should answer. Too much information causes clutter and confusion.

2. Use the right chart type
Choose visualizations that match the data purpose:

- Line charts for trends over time.
- Bar or column charts for comparison.
- Pie or donut charts for part-to-whole relationships (only if few categories).
- Maps for geographical data.
- Cards for key numbers (e.g., Total Sales, Revenue Growth).

3. Apply Gestalt and visual perception principles
Use color, size, and position wisely to direct attention. For example:

- Use one strong color to highlight key metrics.
- Group related visuals close together (principle of proximity).
- Use consistent color and shape for similar data (principle of similarity).

4. Increase data-ink ratio
Remove unnecessary gridlines, borders, and 3D effects. Only keep visual elements that help communicate the data clearly. More ink should represent data, not decoration.

5. Ensure readability
Use sans-serif fonts like Arial or Calibri for text and labels because they are easier to read on screens. Keep font sizes consistent across visuals.

6. Use filters and interactions
Allow users to explore data dynamically with slicers, filters, and tooltips. This makes the dashboard more interactive and useful for decision making.

7. Maintain consistent layout and colors
Keep a consistent color theme for better aesthetics and easy interpretation. For example, use company brand colors for charts and background.

8. Test and refine
Always test your dashboard with end-users to ensure it communicates clearly. A good dashboard tells a story at a glance.

2.5 Machine Learning Integration in BI Model of Loan

In recent years, the rapid advancement of machine learning (ML) has reshaped how financial institutions design and use business intelligence (BI) systems. Instead of relying solely on static dashboards and retrospective reporting, BI platforms are increasingly being connected with ML models to generate predictive insights and support smarter lending decisions.

One of the major contributions in this area comes from Zhu et al. (2023), who combined predictive analytics with explainable AI to enhance credit-risk evaluation. Their study used models like logistic regression, XGBoost, and LightGBM to identify patterns linked to loan defaults, and then visualized the results through interactive BI dashboards. This integration allowed decision-makers not only to detect potential defaulters early but also to understand the reasoning behind model predictions, improving both accuracy and transparency.

Building on the same principle of interpretability, Li (2025) examined nine ML algorithms to determine which models provide the most reliable and explainable outcomes for loan prediction. Instead of treating BI merely as a visualization layer, Li emphasized its role as a communication bridge between complex predictive models and human analysts — translating technical outputs into actionable insights for credit officers.

The work of Mestiri (2024) highlighted how deep learning can be incorporated into BI-based credit scoring systems. By processing large volumes of historical customer and transaction data, the author demonstrated that neural networks and other ML algorithms significantly outperform conventional credit-risk scoring techniques. The study also emphasized the operational advantage of embedding these models into BI platforms, which makes real-time loan evaluation more practical and transparent.

A slightly different approach was taken by Zhang et al. (2025), who developed a full data-driven BI framework for predicting loan defaults. Their system linked data warehouses to an ML layer built with Random Forest and Gradient Boosting models, then pushed the prediction outputs back to a BI environment for visualization. The result was an end-to-end predictive BI solution that continuously refined its accuracy as new loan data became available — a model well-suited for modern, data-intensive banking ecosystems.

Extending the conversation toward deep learning, Chang et al. (2024) integrated ML and neural-network-based models into BI systems to support credit-risk prediction in real time. Their study underscored the importance of automation and adaptive scoring, showing that when BI tools are enhanced by ML, institutions can monitor changing risk levels and respond proactively to early warning signals.

Adding a more analytical perspective, Machado (2025) designed a supervised ML approach that complements BI systems in credit-risk management. Instead of treating analytics as a back-office function, this research positioned ML as an integral layer of BI-driven decision-making — one that helps organizations detect irregularities early and adapt lending policies dynamically.

Taken together, these studies demonstrate a clear shift from descriptive business intelligence toward predictive and prescriptive BI, where machine learning algorithms play a central role. The integration of ML into BI not only boosts prediction accuracy but also bridges the gap between raw data and strategic financial decisions, making loan analytics more transparent, timely, and data-driven.

CHAPTER 3 Requirement Analysis

3.1 Business Problem

In the modern financial landscape, credit risk management plays a decisive role in ensuring the stability and profitability of lending institutions. However, most financial organizations still depend on outdated or semi-manual methods to evaluate loan applicants. These traditional approaches often lack the analytical depth required to forecast potential defaults accurately, leading to rising non-performing loan (NPL) ratios and inefficient capital allocation.

The key business problems that this BI project aims to address are outlined below:

Inefficient Risk Assessment

Current credit evaluation systems rely primarily on manual scoring or static criteria, such as income-to-loan ratio or credit score thresholds. Such methods fail to capture nonlinear patterns and behavioral risk factors that influence a borrower's likelihood to default.

For instance, two borrowers with identical incomes may have different repayment probabilities if their employment stability or existing debt levels differ. This calls for data-driven predictive models integrated into the BI framework to improve risk accuracy.

Fragmented and Non-Standardized Data

Loan and borrower information are often dispersed across multiple systems — including CRM, accounting, and risk databases — without a unified schema. This fragmentation prevents banks from having a 360-degree view of customers. Moreover, inconsistencies in field naming, data types, and entry formats make it challenging to merge datasets into a single analytical model.

Lack of Predictive Intelligence

While existing BI dashboards focus on historical summaries (e.g., default rates by region or by loan purpose), they do not predict future defaults. This reactive approach limits the ability to detect potential risks early and to implement preventive lending strategies.

The integration of machine learning (ML) models into BI systems allows financial institutions to anticipate borrower behavior, assess probability of default, and make preemptive decisions regarding loan approval or restructuring.

Inconsistent Loan Monitoring and Reporting

Without unified, automated dashboards, loan officers must manually prepare reports, which increases human error and delays decision-making. In addition, inconsistency in monitoring standards — especially across different branches or departments — reduces the comparability and reliability of insights.

Strategic Need for Data-Driven Decision-Making

To maintain competitiveness, financial institutions must transition from descriptive reporting to predictive and prescriptive analytics. The goal of this BI project is to build an integrated analytical ecosystem that transforms historical loan data into actionable insights, supporting smarter credit policies and better risk control.

3.2 Data Source

The BI project relies on the **Loan Default Dataset** — a historical dataset of personal loan applications collected from 2013 to 2018. It records borrower demographics, financial attributes, loan characteristics, and repayment outcomes.

The dataset includes the following groups of variables:

a. Quantitative Measures

These are continuous numerical fields used for statistical and predictive modeling:

- **Age:** Borrower's age at loan issuance, indicating life stage and financial maturity.
- **Income:** Annual income, representing financial capacity.
- **CreditScore:** Score from 300–850; a key metric for creditworthiness.
- **LoanAmount:** The total principal borrowed.
- **InterestRate:** Annual rate charged on the loan.
- **LoanTerm:** Duration (in months) for repayment.
- **MonthsEmployed:** Job tenure, reflecting stability.
- **NumCreditLines:** Number of existing credit lines (cards, loans, etc.).
- **DTIRatio:** Debt-to-Income ratio — one of the strongest predictors of loan default.

b. Categorical Attributes

Qualitative characteristics describing the borrower's profile:

- **Education** (High School, Bachelor's, Master's, etc.)
- **EmploymentType** (Full-time, Part-time, Self-employed)

- **MaritalStatus** (Single, Married, Divorced)
- **LoanPurpose** (Home Purchase, Debt Consolidation, Education, etc.)

c. Binary Indicators

Boolean-type variables converted into descriptive categories (“Yes”, “No”, “Unknown”):

- **HasMortgage** — Does the borrower own mortgaged property?
- **HasDependents** — Does the borrower have dependents to support?
- **HasCoSigner** — Is there a co-signer responsible for repayment?
- **Default** — Final outcome variable (1 = defaulted, 0 = repaid).

d. Temporal Information

- **Loan Date:** Represents the origination date (DD/MM/YYYY), allowing trend analysis by time (year, quarter, month).

3.3 Data Challenges

Although the dataset provides a strong foundation for BI analysis, several data-related challenges must be addressed to ensure model reliability and accurate insights:

Data Quality and Completeness

Some fields, such as employment type or education level, contain missing or inconsistent values. Unverified or null entries can distort analysis and model training outcomes. Data cleaning procedures must normalize categorical values and impute missing data where appropriate.

Data Integration

Combining multiple borrower attributes into a cohesive structure presents challenges in aligning dimensions (e.g., mapping binary indicators with descriptive dimensions such as “Yes/No/Unknown”). Consistent surrogate key encoding is required to maintain relational integrity across fact and dimension tables.

Imbalanced Target Variable

In most real-world credit datasets, the number of non-default loans far exceeds defaulted ones. This imbalance can bias machine learning models toward predicting “non-default” more frequently, reducing sensitivity to high-risk cases. Sampling or re-weighting strategies (SMOTE, class weighting) may be necessary to correct this issue.

Feature Correlation and Redundancy

High interdependence between financial variables (e.g., Income, LoanAmount, DTIRatio) can lead to multicollinearity, affecting model interpretability. Feature selection techniques or dimensionality reduction are essential to retain only meaningful predictors.

Scalability and Data Volume

As more loans are processed yearly, the volume of data increases exponentially. BI and ML pipelines must be optimized for scalability, ensuring efficient ETL performance, especially during data refreshes and model retraining.

Analytical and Interpretability Challenges

Financial decision-making requires not only predictive accuracy but also interpretability. Machine learning models must provide explainable insights (e.g., SHAP values, feature importance) to ensure transparency and compliance with regulatory requirements.

3.4 KPIs

In this Business Intelligence (BI) project, a set of Key Performance Indicators (KPIs) is defined to evaluate the performance and risk profile of the loan portfolio from 2013 to 2018. These KPIs serve as the foundation for both descriptive analytics (monitoring trends and distributions) and predictive analytics (identifying risk patterns for default). The main KPIs are categorized into three dimensions: loan volume and growth, financial performance, and credit risk and customer profile.

3.4.1 Loan Volume and Growth Indicators

Total Loan Amount (bn)

○ Definition: Represents the total value of loans issued within the selected timeframe.
○ Purpose: Measures the scale of lending activities and business growth over the years.
○ Visualization: Displayed on the Overview and Loan Amount dashboards as a summary card and line chart.

```
Total Loan Amount = SUM(Loan[LoanAmount])
```

Total Number of Loans

○ Definition: Total number of approved loan applications during the analysis period.
○ Purpose: Indicates lending activity volume and customer engagement.

```
Total Loan = COUNT(Loan[LoanID])
```

YOY Loan Amount Change (%)
○ Definition: Measures the annual percentage change in total loan amount.
○ Purpose: Identifies growth trends and market fluctuations in lending.

```
YOY Loan Amount Change =  DIVIDE( [Total Loan Amount] -
CALCULATE([Total Loan Amount], DATEADD(DimDate[Date], -1, YEAR)),
CALCULATE([Total Loan Amount], DATEADD(DimDate[Date], -1, YEAR)))
```

3.4.2 Financial Performance Indicators

Average Loan Amount (K)
○ Definition: The mean value of all approved loans.
○ Purpose: Provides an overview of average borrower demand.

```
Avg Loan Amount = AVERAGE(Loan[LoanAmount])
```

Average Interest Rate (%)
○ Definition: The mean annual interest rate of issued loans.
○ Purpose: Reflects pricing policies and loan profitability.

```
Average Interest Rate = AVERAGE(Loan[InterestRate])
```

3.4.3 Credit Risk and Customer Profile Indicators

Default Rate (%)
○ Definition: Percentage of loans that defaulted compared to total issued loans.
○ Purpose: Measures credit risk and loan quality performance.

Formula (DAX):

```
Default Rate (%) =

DIVIDE( CALCULATE(COUNTROWS(Loan), Loan[Default] = "Yes"), [Total
Loan])
```

YOY Default Rate Change (%)

- Definition: Year-over-year change in loan default percentage.
- Purpose: Monitors improvements or deterioration in loan quality.

```
YOY Default Rate Change = [Default Rate (%)] - CALCULATE([Default Rate (%)], DATEADD(DimDate[Date], -1, YEAR))
```

3.4.4 Strategic KPIs Summary

Table 3. 1. Strategic KPI Mapping Summary

KPI	Metric Type	Formula (DAX)	Business Objective
Total Loan Amount	Financial	SUM(LoanAmount)	Assess total lending performance
YOY Loan Growth (%)	Growth	YOY Loan Amount Change	Track lending trends
Avg Loan Amount	Financial	AVERAGE(LoanAmount)	Measure average borrower demand
Interest Rate (%)	Financial	AVERAGE(InterestRate)	Evaluate lending profitability
Default Rate (%)	Risk	COUNT(Default="Yes") / COUNT(LoanID)	Identify credit quality
YOY Default Change (%)	Risk	Δ Default Rate by Year	Monitor changes in loan performance

Default by Demographics	Risk/Behavioral	Grouped by Age, Income, Employment, Education, Material Status, Purpose.	Target risk segments
Default by financial characteristic	Risk/Behavioral	Has Signer, Has Mortgage, DTio Ratio, Credit Score.	Target risk segments

3.4.5 Summary

The KPI framework enables a 360° view of lending performance and credit risk. By integrating these metrics into Power BI dashboards, stakeholders can:

- Monitor loan portfolio growth and market trends.
- Detect high-risk borrower segments early using ML predictions.
- Support data-driven credit policy adjustments.

This KPI system forms the analytical foundation for subsequent predictive modeling (Logistic Regression, Random Forest, Gradient Boosting) and visualization in the BI environment.

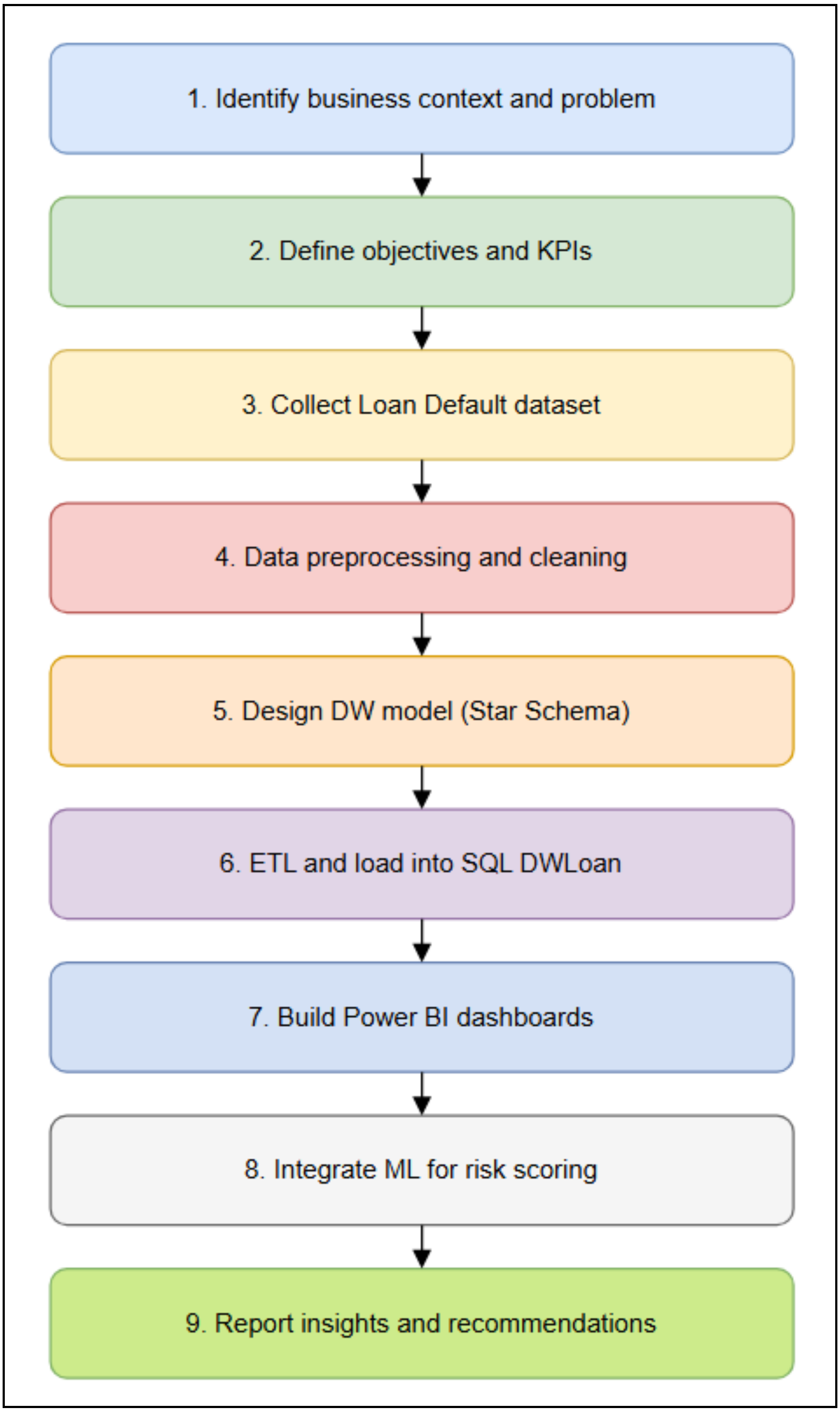

Figure 4. 1. Project Workflow

The development of this Business Intelligence (BI) project follows a structured and data-driven process that integrates techniques from data engineering, visualization, and machine learning. The workflow aims to transform raw financial data into actionable insights that support credit-risk evaluation and decision-making for financial institutions. Each stage in this process builds upon the previous one, ensuring consistency, accuracy, and transparency across the analytical pipeline.

The project began with the identification of the business context and problem definition. The research team examined the operational environment of lending institutions and recognized major challenges such as manual risk evaluation, fragmented data sources, and limited predictive capability. Understanding this context allowed the team to clearly define the project's direction—focusing on automating insight generation and improving risk detection accuracy through BI and analytical modeling.

Following problem identification, the team proceeded to define the project objectives and analytical Key Performance Indicators (KPIs). These KPIs, including Loan Volume, Interest Rate, Default Rate, Credit Score, and Debt-to-Income Ratio (DTIRatio), were selected to measure both financial performance and risk exposure. Establishing these indicators early provided a quantitative foundation for all subsequent data processing, visualization, and modeling tasks.

The next step involved data collection and cleaning. The Loan Default Dataset, covering the period from 2013 to 2018, served as the core data source. It contains demographic, financial, and behavioral information about borrowers, along with loan characteristics and repayment outcomes. The team conducted comprehensive data preprocessing, which included removing missing values, handling outliers, normalizing numerical features, and encoding categorical attributes such as Education, Employment Type, and Loan Purpose. This stage ensured data consistency and readiness for integration into the warehouse.

After ensuring data quality, the team designed the Data Warehouse (DW) using the Star Schema model, a standard architecture for BI systems. The central FactLoan table captures measurable transactions such as Loan Amount, Interest Rate, and Default, while dimension tables (DimDate, DimEducation, DimEmploymentType, DimMaritalStatus, DimLoanPurpose, etc.) provide descriptive context. The design was implemented on Microsoft SQL Server, chosen for its scalability and strong integration with Power BI.

An ETL (Extract – Transform – Load) process was then developed to automate data movement into the SQL-based warehouse. During this process, raw data was extracted from the source files, transformed into a unified format, and loaded into the DWLoan schema. This stage guaranteed data integrity, reduced redundancy, and improved query performance for subsequent visualization and modeling.

Upon successful data integration, the project advanced to data visualization and dashboard development in Power BI. Three main dashboards were created:

- The Overview Dashboard, summarizing key financial and risk indicators;
- The Loan Amount Dashboard, analyzing borrowing patterns across age, income, and loan purpose; and
- The Default Rate Dashboard, focusing on risk segmentation by customer profiles and time periods.

The dashboards were designed based on Gestalt principles, emphasizing clarity, consistency, and interactive exploration through slicers, filters, and dynamic titles.

To enhance analytical depth, the project incorporated Machine Learning (ML) models for predictive risk analysis. Algorithms such as Logistic Regression, Random Forest, and Gradient Boosting were trained to predict default probability based on borrower and loan characteristics. The results were visualized in Power BI, allowing end-users to interpret model outputs and identify high-risk borrowers more intuitively. This integration transformed the BI system from purely descriptive to predictive analytics.

Finally, the project culminated with reporting insights and managerial recommendations. The findings highlighted significant relationships between default risk and factors such as low Credit Scores, high DTIRatio, and unstable employment patterns. The team proposed strategic recommendations for lenders, including revising risk thresholds, prioritizing borrower profiling, and enhancing data governance practices.

4.1 Dataset Introduction

4.1.1 Description of the Loan Default Dataset

The foundational data for this project is sourced from the Loan_default table. This dataset focuses on characterizing loan transactions and the financial profiles of borrowers. The ultimate target variable is the Default indicator, which tracks whether a borrower failed to make timely payments or defaulted on the loan.

4.1.2 Feature Understanding and Data Types

The raw dataset comprises several critical attributes, which can be grouped into identifiers, financial measures, and descriptive categorical attributes:

● Identifiers and Measures: Each entry is uniquely identified by the LoanID. Key quantitative measures include the borrower’s Age, Income, CreditScore (a numerical representation of creditworthiness, typically ranging from 300 to 850), LoanAmount, InterestRate, LoanTerm, and the DTIRatio (Debt-to-Income ratio). The dataset also includes MonthsEmployed and NumCreditLines (active credit lines).

- Categorical Attributes: Descriptive characteristics of the borrower and the loan include Education, EmploymentType, MaritalStatus, and LoanPurpose.
- Binary Indicators: The dataset uses several binary indicators to note the presence of specific conditions: HasMortgage (existing property mortgage), HasDependents (family members to support), and HasCoSigner (someone who shares responsibility for the debt).
- Temporal Data: The date the loan was issued is recorded in the Loan Date (DD/MM/YYYY) field.

Table 4. 1. Raw Data' Columns Definition

Column Name	**Definition**
LoanID	A unique identifier for each loan in the dataset.
Age	The borrower's age at the time the loan was issued.
Income	The borrower's annual income
LoanAmount	The total amount of the loan that the borrower is requesting or has been approved for.
CreditScore	A numerical representation of the borrower's creditworthiness, typically ranging from 300 to 850. A higher credit score indicates the borrower is more likely to repay the debt.
MonthsEmployed	The number of months the borrower has been employed at their current job or with their current employer.
NumCreditLines	The total number of active credit lines (e.g., credit cards, loans) the borrower has at the time of applying for the loan.

InterestRate	The annual percentage rate (APR) charged for borrowing the loan amount, usually expressed as a percentage.
LoanTerm	The length of time (in months) over which the loan is to be repaid.
DTIRatio	The Debt-to-Income ratio, which measures the borrower'debt payments relative to their income. A higher ratio can indicate greater financial stress.
Education	The highest level of education the borrower has completed (e.g., High School, Bachelor's, Master's, etc.).
EmploymentType	The type of employment the borrower is engaged in (e.g., Full-Time, Part-Time, Self-Employed, etc.).
MaritalStatus	The marital status of the borrower (e.g., Single, Married, Divorced, etc.).
HasMortgage	An indicator (e.g., Yes/No) that shows whether the borrower has an existing mortgage on a property.
HasDependents	An indicator (e.g., Yes/No) that shows whether the borrower has dependents (children, other family members) to support.
LoanPurpose	The primary reason for taking out the loan (e.g., Home Purchase, Debt Consolidation, Education, etc.).
HasCoSigner	An indicator (e.g., Yes/No) that shows whether the borrower has a co-signer for the loan (someone who agrees to take responsibility if the borrower defaults).

Default	An indicator (e.g., Yes/No) that shows whether the borrower defaulted on the loan or failed to make timely payments.
Loan Date (DD/MM/YYYY)	The date the loan was issued or originated.

4.2 Data Cleaning and Transformation

The raw data underwent several transformation steps during the Extract, Transform, Load (ETL) process to ensure data integrity and suitability for the dimensional model.

4.2.1 Handling Missing and Outlier Values

- Missing Data in Dimensions: During the population of the dimension tables, any records where the descriptive field (e.g., Education, EmploymentType, MaritalStatus, LoanPurpose) was NULL were explicitly filtered out.
- Unknown Indicator Values: For the binary indicator fields (HasMortgage, HasDependents, HasCoSigner), which are assumed to hold numerical values (0 or 1) in the source, a specific transformation was implemented. Any value other than 0 or 1 was mapped to the descriptive category 'Unknown' within the dimension tables, while 1 was mapped to 'Yes' and 0 to 'No'.

4.2.2 Data Normalization and Encoding

The primary normalization technique applied was the Dimensional Modeling process itself, separating descriptive attributes from measures.

- Surrogate Key Encoding: Textual categorical fields were converted into Surrogate Keys (e.g., EducationKey, MaritalStatusKey). These integer keys, assigned via an identity property (IDENTITY(1,1)), serve as the primary keys for all dimension tables and foreign keys in the fact table.
- Date Encoding: The source date field was transformed into the DateKey. This key is an integer representation of the date, standardized in the YYYYMMDD format, allowing for efficient time-based querying.

4.2.2.1 SQL Transformations and Business Rules

The transformation layer applied various SQL logic to enforce business requirements:

1. Schema Creation: All resulting DW tables were organized under a dedicated schema named DWLoan.

2. Date Attribute Extraction: The date field was parsed to extract granular temporal attributes, including Year, Quarter, Month, and Day, which were stored in the DimDate table alongside the FullDate.
3. Dimension Population Logic: Each dimension was populated using a SELECT DISTINCT query against the raw data to ensure that only unique descriptive values were inserted, receiving their own unique surrogate key.

4.3 Data Warehouse Construction

4.3.1 Designing the Star Schema

The Data Warehouse was constructed based on the Star Schema principle, optimizing for query performance and analytical flexibility.

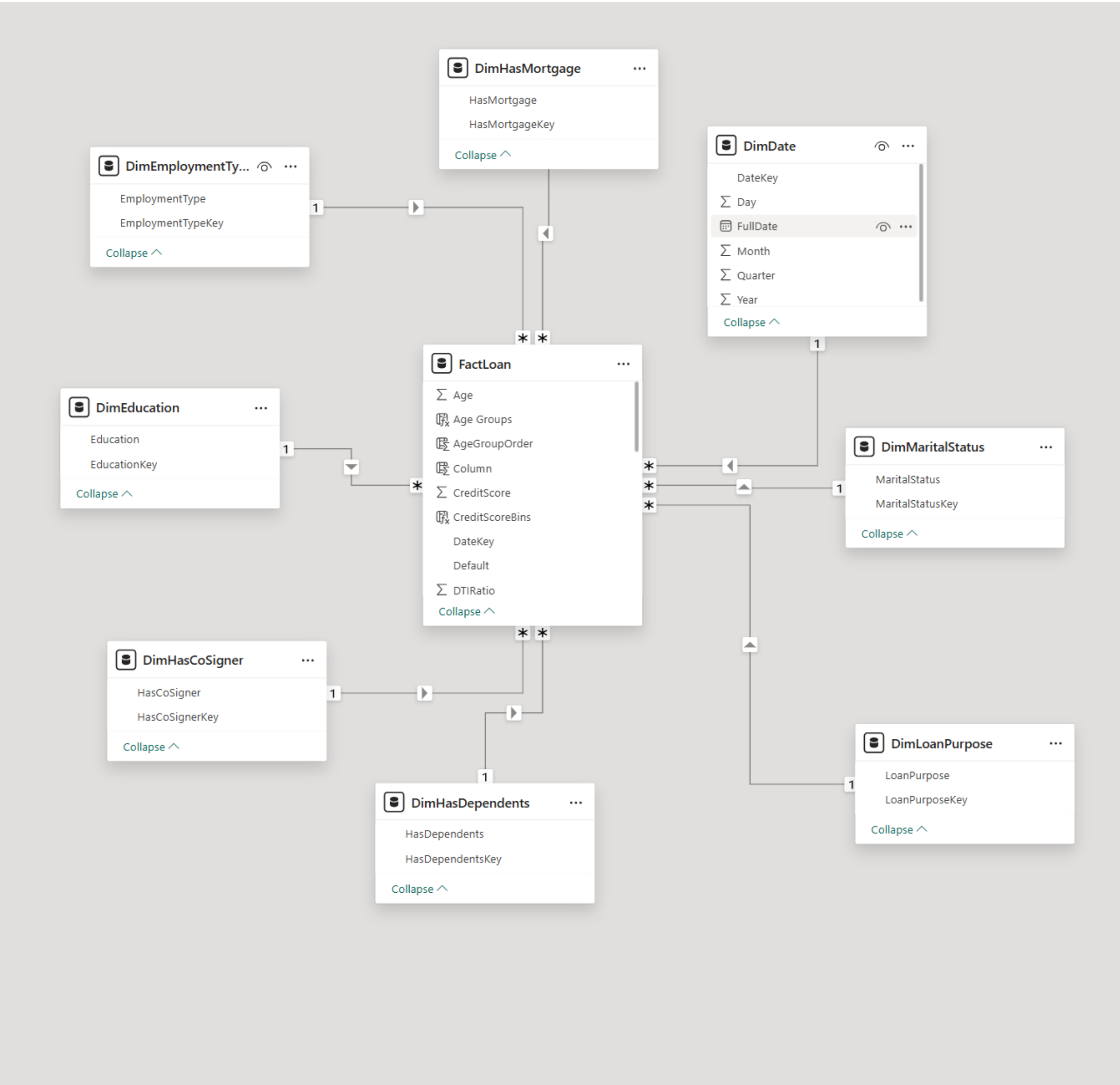

Figure 4. 2. Loan DW Model

a. Fact Table: Loan Fact

The central table is DWLoan.FactLoan.

- Key Structure: It utilizes a FactLoanKey as its primary key. It contains foreign keys linking to all dimension tables, including EducationKey, EmploymentTypeKey, DateKey, HasMortgageKey, HasDependentsKey, and HasCoSignerKey.
- Fact Measures: This table houses all quantitative and additive measures directly related to the loan event: Age, Income, CreditScore, MonthsEmployed, NumCreditLines, LoanAmount, InterestRate, LoanTerm, and DTIRatio. The outcome variable, DefaultFlag (stored as a BIT data type), is also included here.

b. Dimension Tables: Borrower, Time, Loan Purpose:

All dimension tables utilize surrogate keys created with the IDENTITY(1,1) property.

1. Time Dimension: The DWLoan.DimDate table stores the descriptive details of the loan origination date, indexed by the integer DateKey (YYYYMMDD).
2. Loan Purpose Dimension: DWLoan.DimLoanPurpose stores the distinct reasons for the loans.
3. Borrower Demographic/Status Dimensions: Attributes describing the borrower's background are stored in separate, distinct tables:
    - DWLoan.DimEducation.
    - DWLoan.DimEmploymentType.
    - DWLoan.DimMaritalStatus.
4. Indicator Dimensions: The three binary indicators were modeled as separate dimensions to facilitate filtering based on 'Yes', 'No', or 'Unknown' statuses:
    - DWLoan.DimHasMortgage.
    - DWLoan.DimHasDependents.
    - DWLoan.DimHasCoSigner.

4.3.2 Data Loading into SQL Server DW

The final data loading phase involved joining the raw data source (loan.dbo.Loan_default) with the newly created dimension tables to retrieve the appropriate surrogate keys.

The FactLoan table was loaded by executing a complex SELECT statement that performed inner joins across all nine dimension tables (e.g., DimEducation e, DimHasMortgage hm, etc.). The join conditions matched the descriptive attributes (e.g., e.Education = src.Education) to successfully retrieve the corresponding foreign keys (e.EducationKey).

CHAPTER 5 Data Visualization in Power BI

Chapter 5 focuses on the visualization and analytical interpretation of loan data through interactive dashboards built in Microsoft Power BI. This chapter serves as the bridge between data preparation and insight generation, transforming the cleaned and integrated datasets stored in the Data Warehouse into meaningful business intelligence. By leveraging Power BI's robust visualization capabilities, this stage enables decision-makers to monitor key performance indicators (KPIs), identify emerging trends, and evaluate borrower behavior and credit risk in an intuitive, data-driven manner.

The primary objective of this chapter is to translate complex financial and demographic information into accessible visual insights. Through a series of dashboards, the chapter illustrates the overall loan portfolio performance, explores borrower characteristics such as age, income, education, and employment type, and highlights default risk patterns associated with loan amount, interest rate, and credit score. Each visualization was carefully designed based on principles of effective data storytelling, ensuring clarity, consistency, and analytical depth while maintaining a user-friendly layout that facilitates both strategic overview and detailed drill-down analysis.

In addition, this chapter demonstrates how interactive features such as filters, slicers, bookmarks, and drill-through actions enhance analytical flexibility and user engagement. These elements allow business users to dynamically explore relationships within the dataset, uncover hidden correlations, and derive actionable insights without requiring advanced technical expertise. The integration of Power BI with the Data Warehouse ensures that the dashboards remain scalable, maintainable, and refreshable, supporting continuous decision-making in real time.

Ultimately, the goal of Chapter 5 is not only to present visual results but also to establish a framework for data-driven financial analysis. Through the systematic visualization of key indicators such as loan amount, interest rate, and default behavior, the dashboards provide a comprehensive foundation for evaluating the performance and risk exposure of the lending portfolio. This visual intelligence will serve as the analytical baseline for Chapter 6, where predictive modeling and advanced machine learning techniques are introduced to further quantify credit risk and optimize lending strategies.

Master Data (Introduction of the Report):

Figure 5. 1. Master Data Page

The first page serves as the introductory interface of the project, providing an overview of its objectives, scope, and analytical framework. This page outlines the context of the Report. It highlights the project's mission to support financial institutions in identifying high-risk borrowers, improving credit assessment, and minimizing default-related losses. The visual layout also features three interactive navigation buttons—Overview, Loan Amount, and Default Rate—which guide users to detailed analytical dashboards. These buttons not only enhance the user experience but also create a structured analytical journey across the report. Overall, this introductory page functions as both an executive summary and a navigation hub, bridging the conceptual background of the study with its data-driven exploration in subsequent pages.

5.1 Overview Dashboard

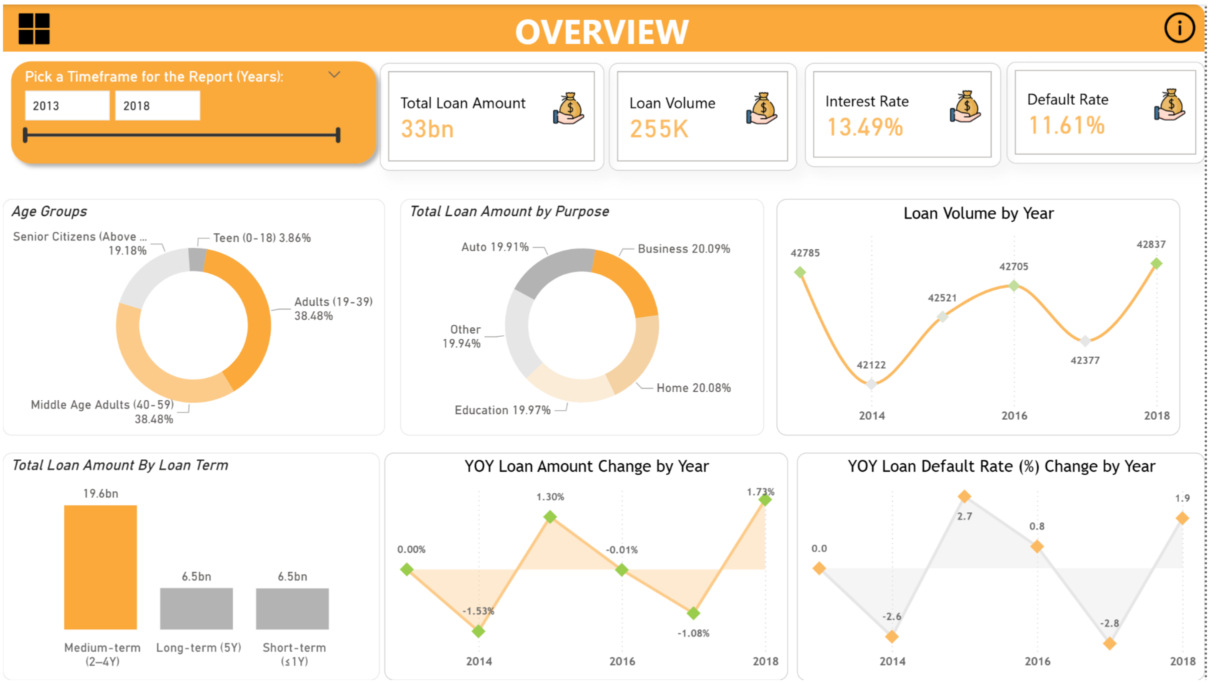

Figure 5. 2. Overview page.

The Overview page serves as the overview interface of the reporting system, designed to provide a comprehensive view of the business's lending activities from 2013 to 2018. The goal of this page is to help users quickly grasp the general situation of the loan portfolio size, customer structure, main credit product groups, as well as the fluctuation trends of financial indicators and risks over time. This is where general information is gathered before users delve into detailed analysis pages such as Loan Amount, Interest Rate, or Default Risk.

The layout of the page is divided into three main sections: the KPI Summary, the Portfolio Composition, and the Trend Analysis. Each section plays a separate role in supporting analysis and providing an overview of the organization's credit performance.

The KPI Summary section, located at the top of the page, displays four key indicators that reflect the performance and risk profile of the loan portfolio. These indicators include: Total Loan Amount, which represents the size of the credit capital that the organization has disbursed over the entire period; Total Loan, which reflects the volume of credit transactions and the level of customer expansion; Average Interest Rate, which shows the average profitability of loans; and Default Rate, which shows the level of credit risk in the portfolio. These four indicators are arranged evenly, using consistent colors and intuitive illustrative icons, making it easy for viewers to grasp the situation with just a general view. Through this section, users can immediately assess the size, performance, and risk of the loan portfolio for each year or the entire analysis period.

The second section, Portfolio Composition, focuses on describing the loan portfolio structure according to demographic characteristics and credit types. Three main charts are presented in this area: Age Groups, Loan Purpose, and Loan Term. The age group chart allows viewers to identify the main customer group, thereby assessing the concentration or diversity in the borrower. The loan purpose chart shows the proportion of capital allocated to product types such as home loans, business loans, education loans, or personal loans, thereby helping to determine the strategic direction and key market segments of the organization. Finally, the loan term chart reflects the borrower's tendency to choose the term as short-term, medium-term, or long-term, thereby helping to assess the risks related to the payback period and credit cash flow. All three charts are designed in donut chart or bar chart format, using the same orange and gray tones to create visual consistency and emphasize the relationship between the elements that make up the loan portfolio.

The last section of the page, Trend Analysis, is used to track and evaluate lending trends over the years. Three main charts are located in this area: Loan Volume by Year, YOY Loan Amount Change (% Change by Year), and YOY Loan Default Rate Change (% Change by Year). The first chart shows the number of loans over the years, allowing you to identify periods of growth or decline in credit activity. The second chart shows the percentage change in the total loan value from year to year, reflecting the growth or contraction in lending. The third chart shows the fluctuation in default rates over the years, helping to detect periods of unusually high risk. These three charts combine to create a logical story about the relationship between growth, profit, and risk, allowing users to monitor the performance of the organization over time in a visual and systematic way.

In addition, the Overview page also integrates a Year Range Slicer that allows users to customize the analysis period from 2013 to 2018. The charts are all interactive and linked together, making it easy for users to drill down or cross filter to see details by age group, loan purpose, or specific term.

The four KPIs at the top of the dashboard: Total Loan Amount, Total Loan, Interest Rate, and Default Rate serve as core measures that reflect the organization's credit performance over the 2013–2018 period.

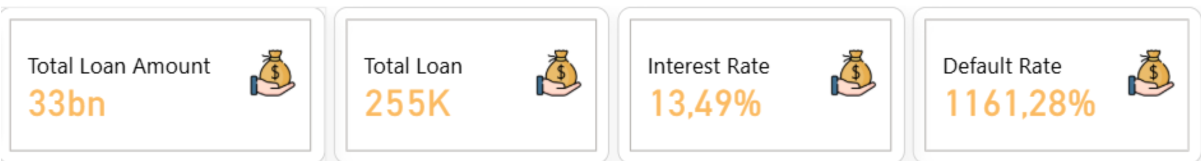

Figure 5. 3. KPI: Overview page KPIs

Total Loan Amount reached 33 billion, representing the total value of loans disbursed during the entire analysis period. Reaching the total value of 33 billion shows that the loan portfolio has reached a large enough scale to reliably assess trends and risk structure.

Total Loan is 255K, representing the number of loans issued during the entire period. This is an indicator that reflects the level of activity and frequency of credit transactions of the organization. Having more than 255,000 loans shows a high level of portfolio diversification, good risk distribution among many customer groups. When combining the first two indicators, we see that although the number of loans is large, the total loan value is only average, which implies that the majority of loans are small or medium in size, in line with the strategy of targeting individual customer groups or small and medium-sized enterprises. This is a positive sign, as it helps to minimize the risk of capital concentration in a few large customers.

The third indicator, Interest Rate = 13.49%, reflects the average interest rate that the organization applies to loans. This interest rate is considered relatively high compared to the general level, indicating that the business is pursuing an aggressive risk pricing strategy, or is operating in a customer segment with higher credit risk than average, such as individuals with low Credit Scores or short-term consumer loans. From a financial perspective, a high interest rate increases profit margins, but at the same time, it also has the potential to increase the default rate if the customer's ability to repay is not commensurate. Therefore, this indicator should be considered in parallel with the Default rate to assess the balance between profit and risk.

With a Default Rate of 11.61%, a level of 11–12% means that about 1 in 9 loans is in default or does not perform its obligations on time. This is not too low, especially if the portfolio is aimed at individual customers and SMEs. This number indicates that the business accepts a relatively high level of risk in exchange for high interest rates, meaning that the business strategy is oriented towards growth and profit rather than fully optimized for credit safety.

Overall, when we put the four indicators side by side, we see a fairly consistent picture: large scale, high interest rates, and relatively high risk.

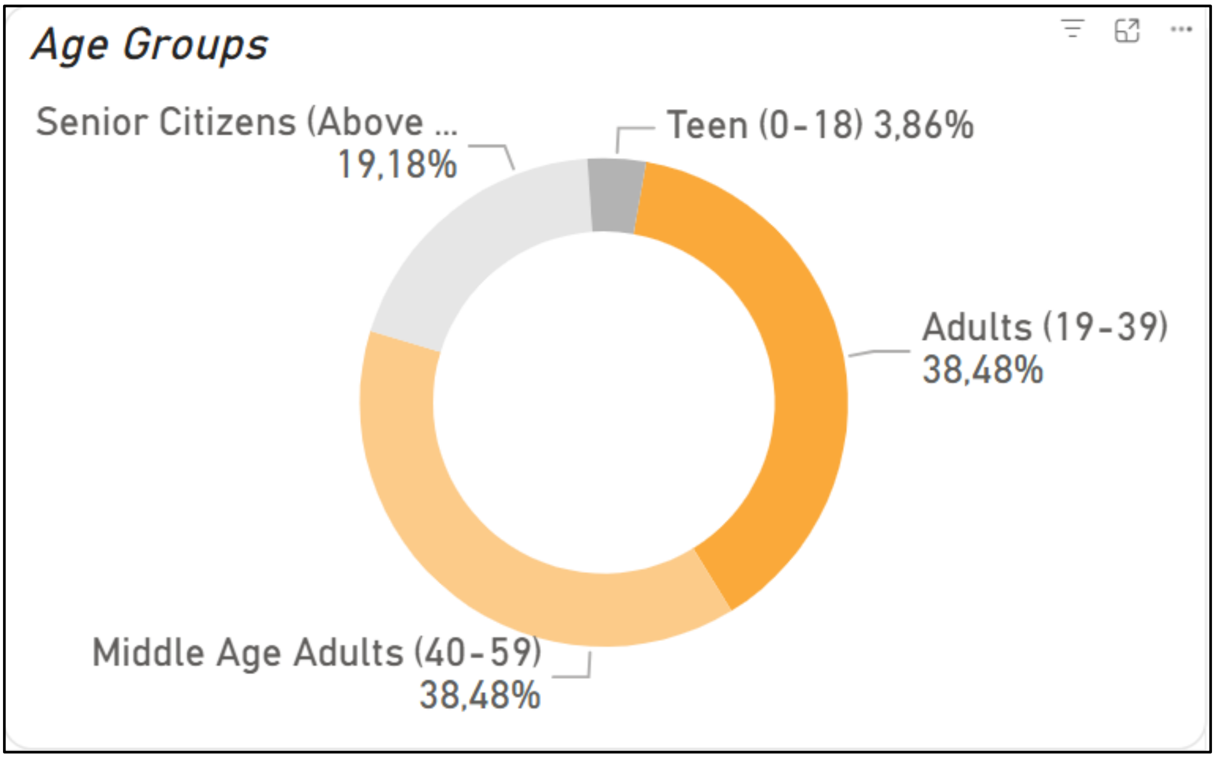

Figure 5. 4. Age Groups

First of all, the donut chart Age Groups shows that the majority of borrowers are in the two age groups 19–39 (38.48%) and 40–59 (38.48%), accounting for a total of nearly 77% of the entire portfolio. These are groups in their prime working age, with stable incomes and high credit needs for consumption, housing, or personal investment. This accurately reflects the business's strategy of targeting dynamic customers who are in the stage of personal economic development. The Senior Citizens group (over 59 years old) accounts for about 19.18%, while the Teen group (0–18 years old) is only about 3.86%, showing that credit products for young people or retirees still account for a small proportion. This structure is relatively reasonable because the business is focusing on the customer group with good debt repayment ability and the strongest demand for consumer or investment loans.

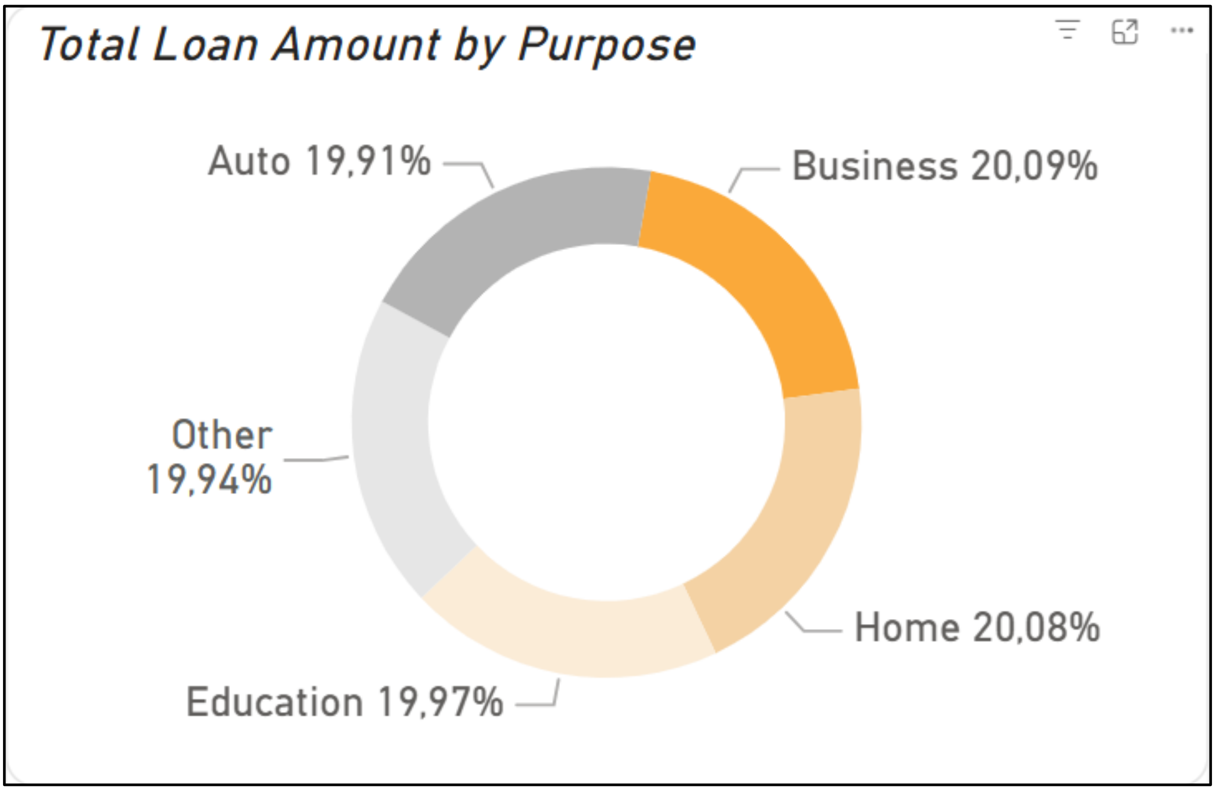

Figure 5. 5. Total Loan Amount by Purpose

Next, the donut chart Total Loan Amount by Purpose shows that the loan portfolio is quite evenly distributed among the following purpose types: Business (20.09%), Home (20.08%), Education (19.97%), Auto (19.91%), and Other (19.94%). This structure shows that the enterprise's loan portfolio is highly diversified, not dependent on a single sector. This is both a strength in terms of risk management and reflects the enterprise's strategic orientation in simultaneously exploiting many credit market segments. However, the absolute level of uniformity between groups may also suggest that the enterprise does not have outstanding specialized products, which could be an opportunity to develop further in some highly profitable segments such as "Business Loans" or "Education Loans" if risks are well controlled.

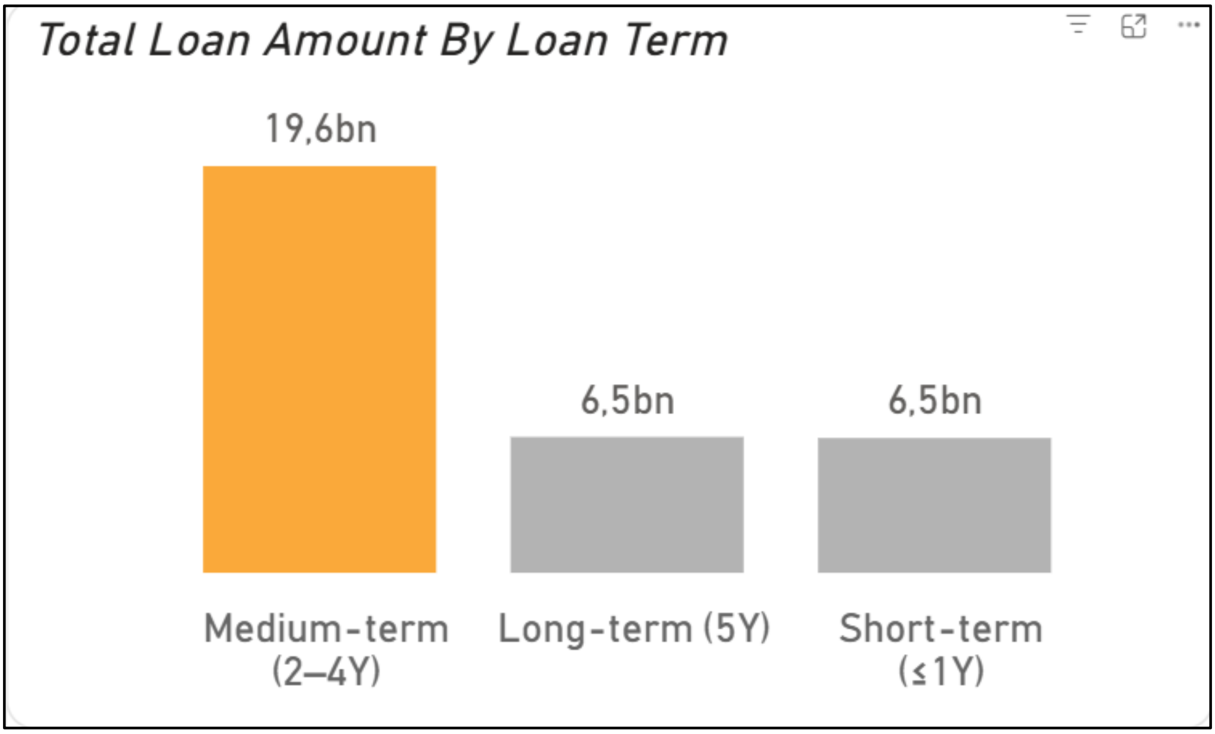

Figure 5. 6. Total Loan Amount By Loan Term.

The column chart Total Loan Amount by Loan Term shows that medium-term loans (2–4 years) are absolutely dominant at around 19.6 billion, while short-term (<1 year) and long-term (5 years and above) are both around 6.5 billion. This pattern suggests that businesses prefer loan structures with moderate repayment cycles, balancing profits and liquidity risks. Medium-term loans typically offer higher interest margins than short-term loans but still ensure better risk management than long-term loans. This distribution suggests a relatively cautious credit policy aiming for sustainable growth rather than massive expansion.

The current loan portfolio structure is considered stable and reasonable, focusing on the customer group with the best payment ability and the most popular loan products. However, to enhance competitive advantage, businesses can consider developing specialized products according to each age group or loan purpose to expand market share and optimize profits. The current portfolio structure reflects a healthy credit foundation, but also shows untapped potential in emerging and long-term segments.

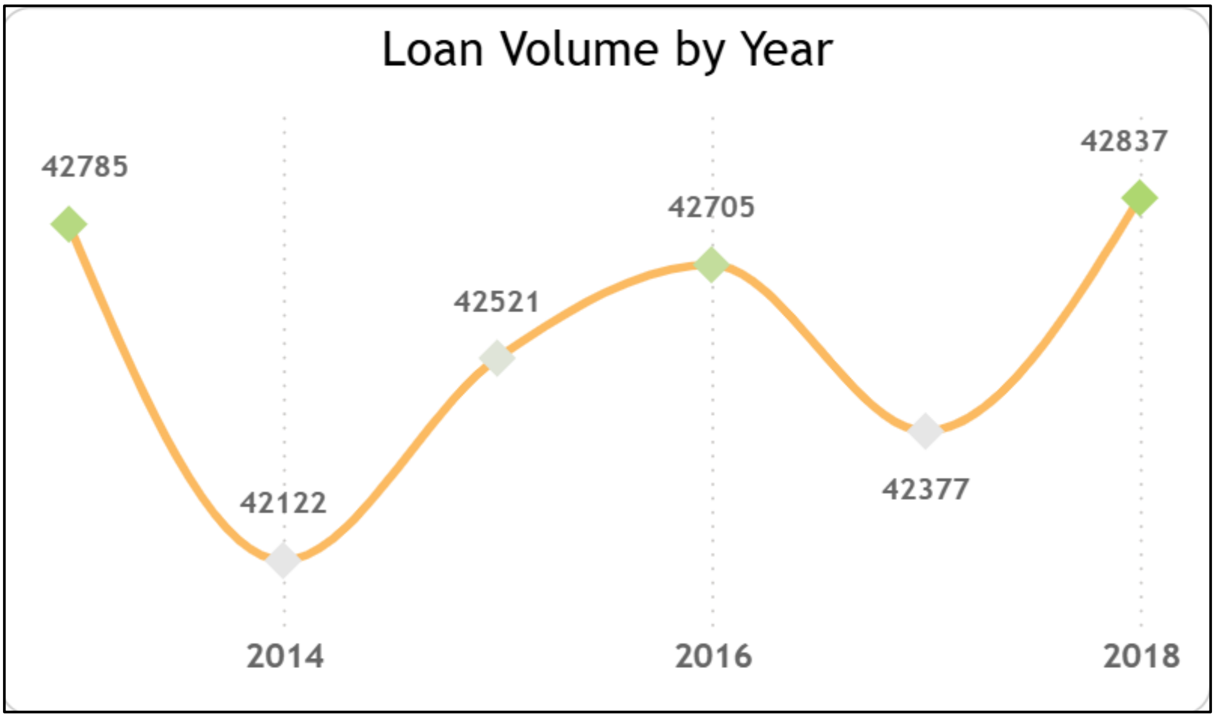

Figure 5. 7. Loan Volume by Year chart.

Loan Volume by Year is a line chart in which the X-axis represents the years from 2013 to 2018, while the Y-axis represents the number of loans issued. Overall, the line shows a trend of alternating fluctuations between years, rather than a continuous increase. Specifically, the loan volume decreased slightly from 42,785 in 2013 to 42,122 in 2014, then recovered and peaked at 42,705 in 2016, then declined again before increasing sharply again in 2018, reaching 42,837. This fluctuation pattern reflects the cyclical nature of credit activities, which can be influenced by external factors such as monetary policy, macroeconomic conditions, or temporary credit tightening. However, overall, the performance curve still maintains a slight upward trend, showing that businesses have maintained stability in lending activities, despite cyclical fluctuations.

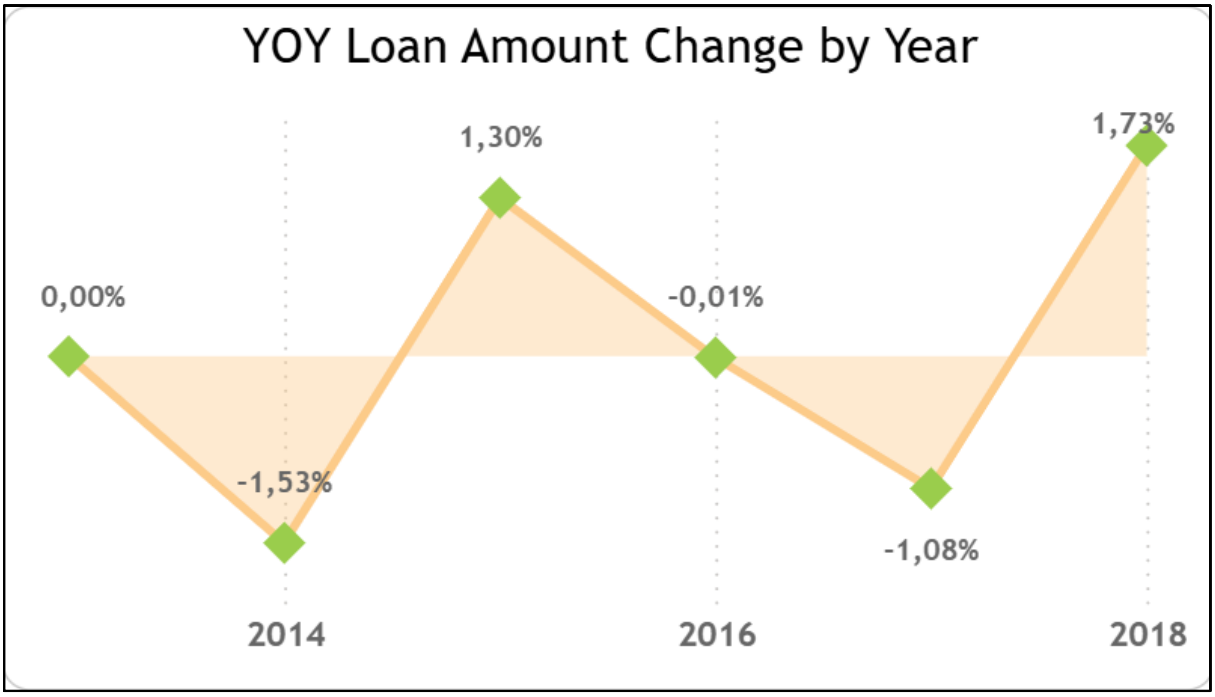

Figure 5. 8. YOY Loan Amount Change by Year Chart

YOY Loan Amount Change by Year, also represented by a line chart, in which the X-axis is the year, and the Y-axis represents the percentage change in the total loan amount compared to the previous year. This chart helps determine the annual credit growth rate. The data shows that 2015 recorded the strongest growth rate of +1.30%, while 2014 and 2017 both had negative values, at -1.53% and -1.08%, respectively. After a period of downward adjustment, in 2018, the growth index reached +1.73%, the highest in the entire period. This shows that the credit activities of enterprises did not increase steadily, but went through distinct cycles of expansion, contraction, and recovery.

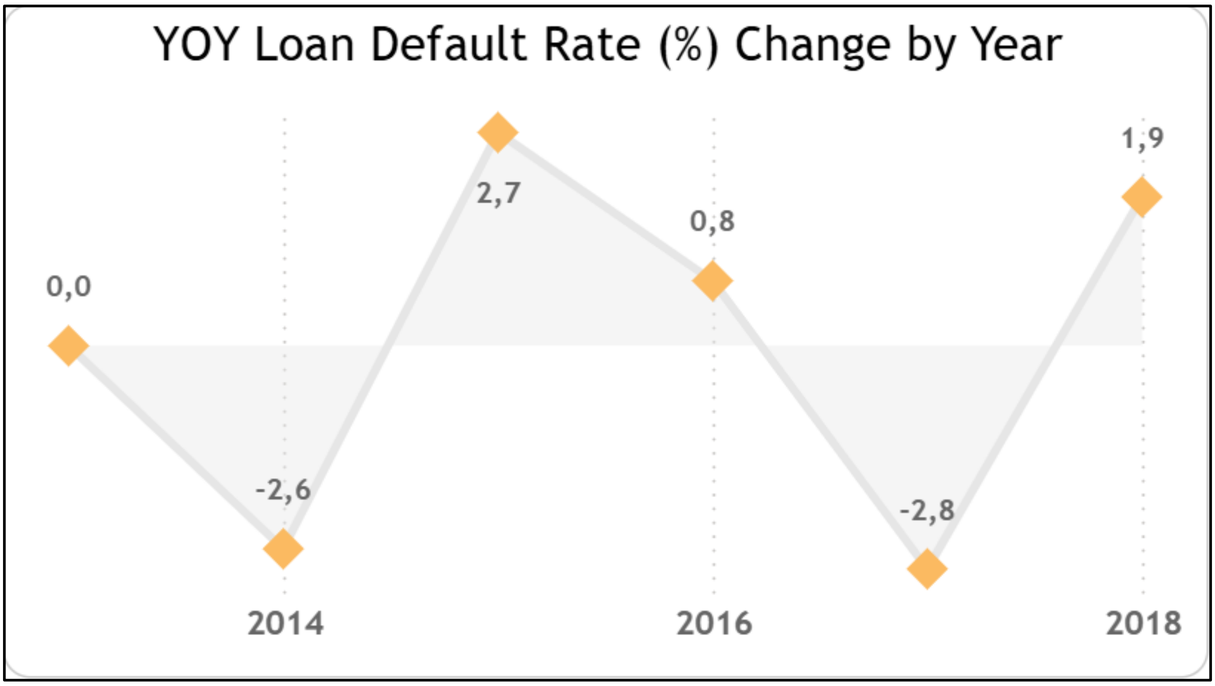

Figure 5. 9. YOY Loan Default Rate (%) Change by Year chart.

While the above two charts focus on growth, the third chart, YOY Loan Default Rate (%) Change by Year, reflects the opposite direction of credit activity, i.e., bad debt risk. This is also a line chart, in which the X-axis represents time and the Y-axis represents the percentage change in the default rate compared to the previous year. The line shows strong fluctuations in the default rate, often in the opposite direction of credit growth. In 2014, when businesses reduced their lending volume, the default rate dropped significantly to -2.6%. However, in 2015 and 2016, the period of credit expansion, the default rate also increased, reaching 2.7% and 0.8%, respectively. By 2017, this ratio had fallen sharply to -2.8%, then rebounded to 1.9% in 2018. This pattern shows a fairly clear relationship between credit expansion and credit risk: when businesses increase lending, bad debts tend to increase; when policies are tightened, risks immediately decrease.

It can be seen that during the period 2013-2018, the enterprise experienced short-term credit fluctuations but still maintained a stable growth trend. The positive growth in loan volume in 2018 along with good control of default rate shows that the enterprise not only expanded its scale of operations but also maintained the quality of its credit portfolio. This structure of fluctuations between growth and risk shows a reasonable balance between profit and financial safety, proving that the enterprise has achieved a certain level of maturity in its credit management strategy.

5.2 Loan Amount Page

After the Overview page has provided an overview of the organization’s lending activities, including size, growth rate, and credit risk, the Loan Amount page is built to analyze the structure and dynamics of total loan value in the period 2013–2018. The goal of this page is to clarify the factors that directly affect the total loan amount, such as time, customer group, type of occupation, education level, income, and loan purpose. Understanding these factors helps determine which customer group brings the largest loan value, which loan purpose accounts for the highest proportion, and what conditions are affecting borrowers’ credit behavior.

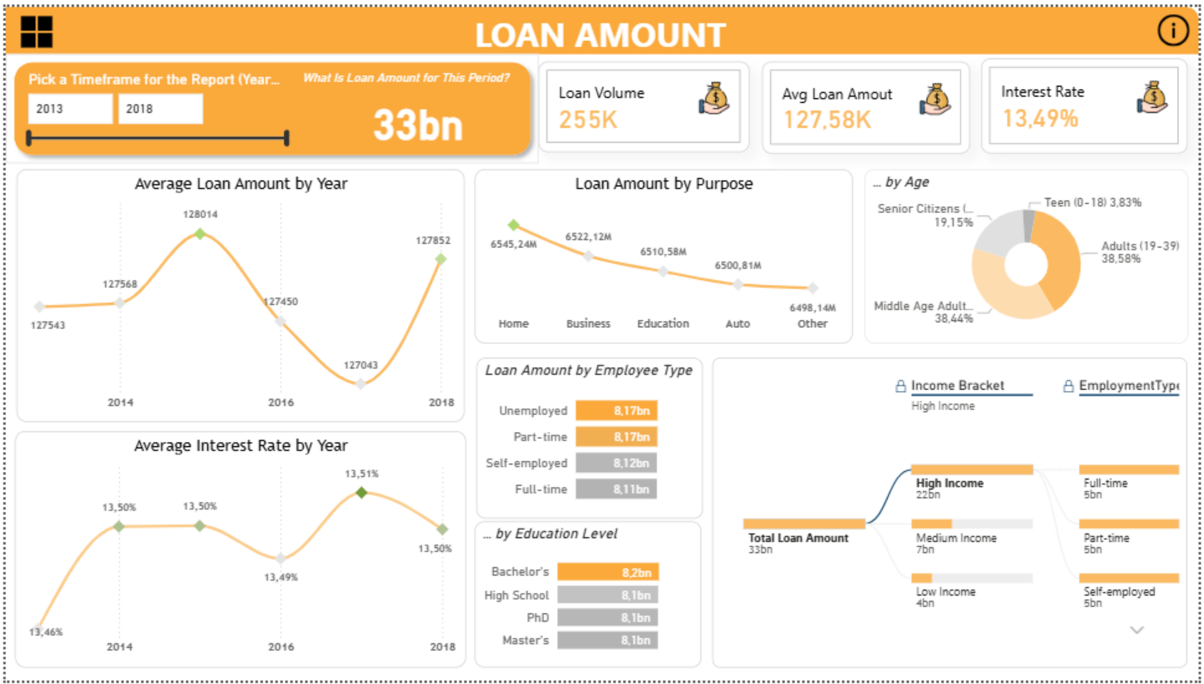

Figure 5. 10. Loan Amount Page

In addition, the Loan Amount page also plays an important role in evaluating the effectiveness of a business's credit strategy. While the Overview page focuses on the balance between growth and risk, this section shifts the focus to the quality of growth. Clarifying this structure not only helps understand the current performance of the loan portfolio, but also serves as a basis for guiding future product development and credit risk management policies.

The Loan Amount page is designed with a balanced and clearly stratified layout, ensuring that readers can receive information from overview to detail in an intuitive and logical way. The top of the page is a cluster of summary indicators displayed right below the title, including Total Loan Amount, Loan Volume, Average Loan Amount, and Interest Rate. Below the KPI section, the page is divided into three main content groups, each group showing a different aspect of analysis: fluctuations over time, analysis of portfolio structure, and analysis of demographics, occupations, and income.

The first group consists of two line charts: Average Loan Amount by Year and Average Interest Rate by Year. Both charts use the horizontal axis (X-axis) to represent the time period from 2013 to 2018, while the vertical axis (Y-axis) represents the average loan value and the corresponding interest rate. Placing these two charts side by side allows readers to easily compare the fluctuation trends, thereby detecting the relationship between interest rates and customers' borrowing behavior over the years.

The second group focuses on analyzing the structure of the loan portfolio, shown through the Loan Amount by Purpose chart. This chart uses a line chart or column chart, in which the X-axis represents the loan purposes such as Home, Business, Education, Auto and Other, while the Y-axis represents the total loan value by each group. This way of presentation helps to clearly identify the areas that account for the highest proportion of the credit portfolio. Next to it is the Loan Amount by Age chart in donut form, illustrating the contribution rate of each age group such as Teen, Adults, Middle Age Adults and Senior Citizens. The combination of these two charts helps to form a parallel view of loan product characteristics (purpose) and borrower characteristics (age), creating a foundation for the analysis of borrowing behavior in the following paragraphs.

The third group delves into demographic and occupational factors, with the Loan Amount by Employee Type and Loan Amount by Education Level charts in horizontal bar form. These charts allow for direct comparison between groups of workers (full-time, part-time, self-employed, unemployed) and education levels (High School, Bachelor's, Master's, PhD) in terms of loan size. Notably, in the same group is also the Decomposition Tree showing the relationship between Income Bracket and Employment Type.

5.2.1 Dashboard analysis

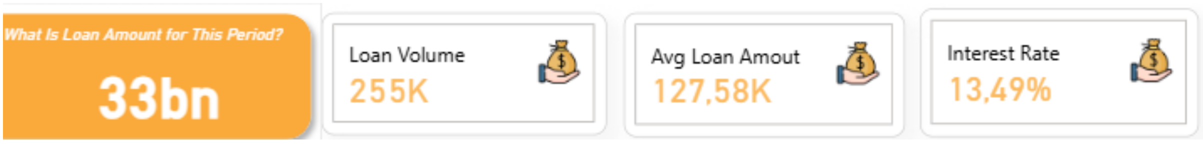

Figure 5. 11. Loan Amount KPIs.

The total loan value (VND 33 billion) shows the scale of operations, while the loan volume (255 thousand) reflects the openness of the market and the level of customer access. The average loan value (127.58 thousand) helps to assess the average level of personal credit, while the interest rate (13.49%) reflects the current financial policy. When combining these four indicators, we can visualize the overall picture of the scale, quality, and cost of credit, which will serve as a basis for more in-depth analysis in the following sections.

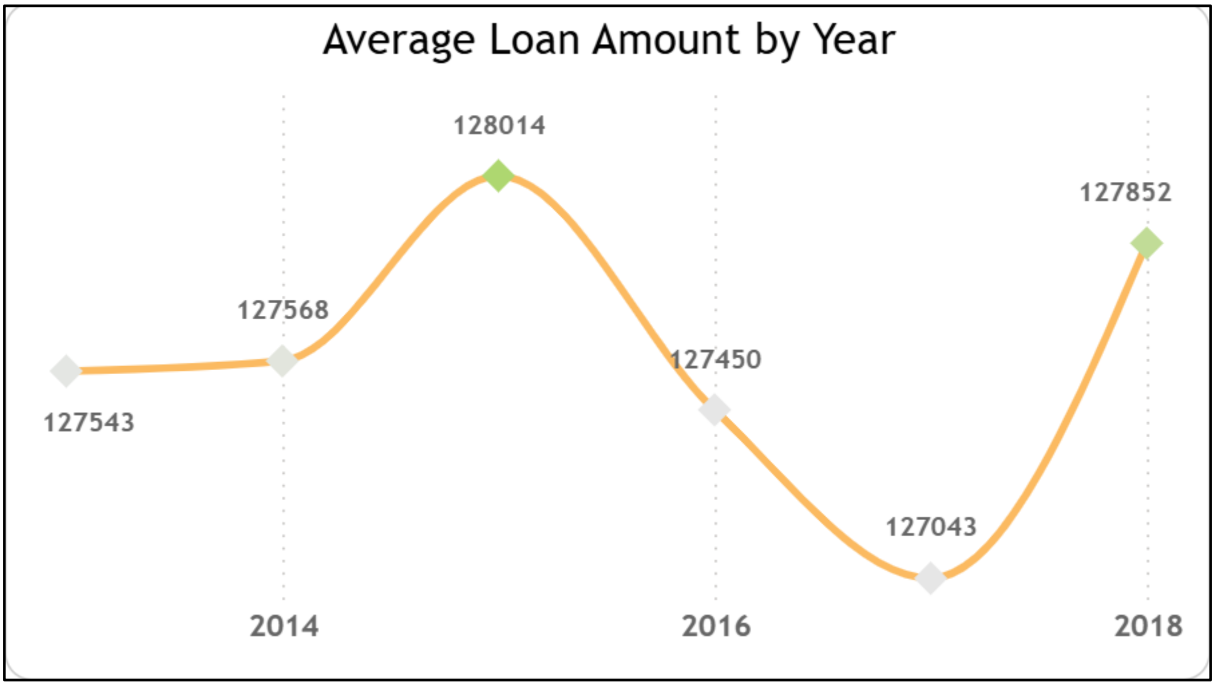

Figure 5. 12. Average Loan Amount by Year.

The Average Loan Amount by Year chart is presented in the form of a line chart, in which the horizontal axis (X-axis) represents the time series from 2013 to 2018, and the vertical axis (Y-axis) represents the average value of loans in each year. This chart helps readers observe the trend of fluctuations in the average loan value over the years, reflecting changes in personal credit demand and business lending policies. Observing the performance line shows that the average loan value is not completely stable but fluctuates slightly over the periods. In the first period, from 2013 to 2015, the loan value increased steadily and reached its highest level in 2015, about more than 128 thousand VND per loan, showing a period of strong credit expansion. However, from 2016 to 2017, the average value decreased slightly, possibly due to tightening lending policies or fluctuations in macroeconomic conditions. By 2018, the average value recovered, reaching about 127.8 thousand VND, marking the stability of the loan portfolio after the adjustment period. The small but steady fluctuations show that businesses maintain a stable loan structure, not too affected by economic fluctuations. However, the slight decrease in 2017 is also a signal that the market may be sensitive to factors such as changes in interest rates or credit conditions. Therefore, businesses need to continue to monitor this trend in the long term to ensure a balance between access to capital and the safety of the loan portfolio.

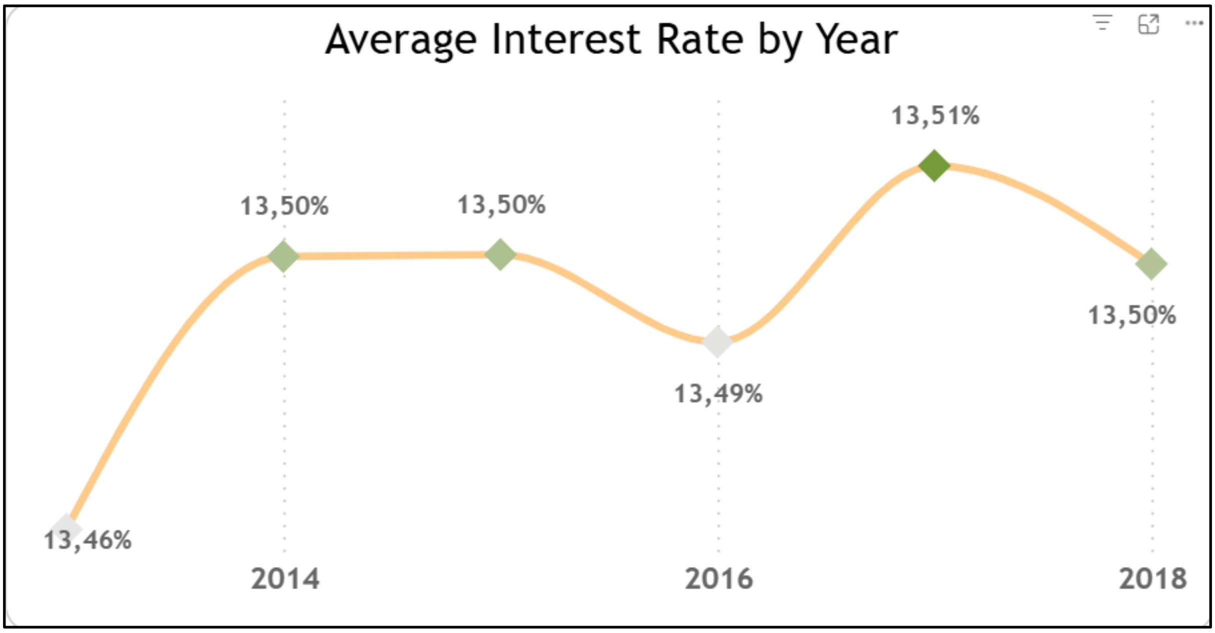

Figure 5. 13. Average Interest Rate by Year chart

The Average Interest Rate by Year chart is also presented as a line chart, with the same X-axis as time and the Y-axis as the average interest rate of loans in each year. This chart is placed alongside the Average Loan Amount by Year chart for comparison purposes, thereby understanding the relationship between the interest rate and the average loan amount. The average interest rate has been remarkably stable throughout the observation period, fluctuating around 13.4% to 13.5%, only increasing slightly in 2017. This stability shows that businesses maintain a flexible interest rate policy and control credit costs well, without putting too much pressure on borrowers. Notably, the period of 2017, when interest rates increased slightly, coincided with the period of average loan value decreasing in the previous chart. This shows an inverse relationship: when borrowing costs increase, borrowers tend to reduce the size of their loans, reflecting the sensitivity of the credit market to interest rate fluctuations.

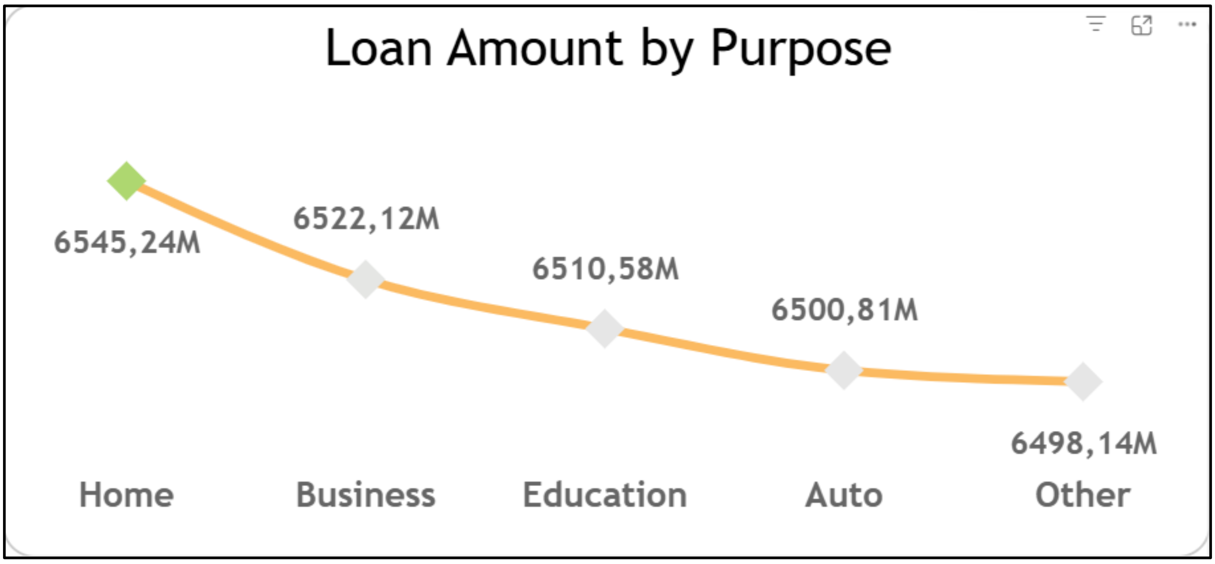

Figure 5. 14. Loan Amount by Purpose chart

The Loan Amount by Purpose chart is presented in the form of a column chart, in which the horizontal axis shows the loan purpose groups including Home, Business, Education, Auto and Other, while the vertical axis shows the total loan value of each group. Observing the data shows that the five loan purpose groups have relatively balanced values, fluctuating around 6.4-6.5 billion VND, in which the Business and Home groups are slightly higher. This structure reflects the high diversification in the loan product portfolio, showing that businesses do not depend on a single type of credit but divide the risk equally among many different fields. The fact that the Education and Auto groups have similar values also shows that the personal credit market is increasingly expanding in scope, aiming at many other spending goals besides traditional housing or business. Therefore, in the coming period, the organization can consider reallocating portfolio weights towards prioritizing products with higher growth potential and profit margins, while continuing to maintain diversification as a foundation for risk reduction.

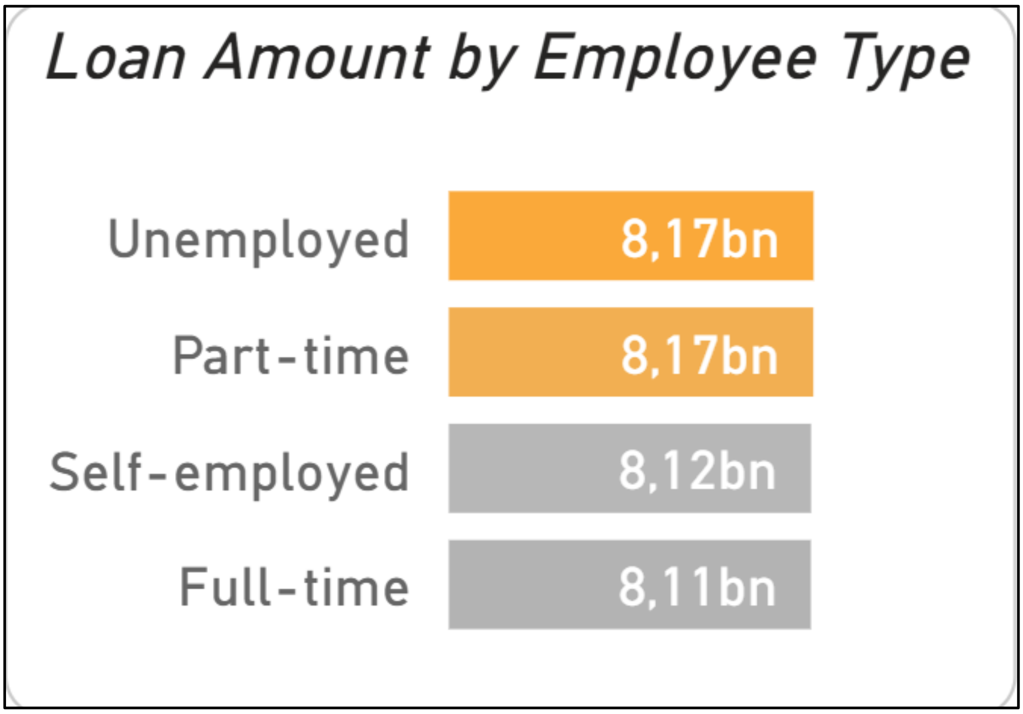

Figure 5. 15. Loan Amount by Employee Type

The Loan Amount by Employee Type chart is presented as a horizontal bar chart, in which the vertical axis shows the groups of employment types including Full-time, Part-time, Self-employed, and Unemployed, and the horizontal axis shows the total value of the corresponding loans. Observing the chart shows that the occupational groups have quite similar loan values, fluctuating around 8.1–8.2 billion VND, in which the Full-time and Self-employed groups often have a slightly higher value than the Part-time and Unemployed groups. The nearly equivalent loan levels between groups show that the business is applying a fairly flexible credit policy, not creating significant barriers between customers, including the group of freelancers or those without stable jobs. However, the small but consistent differences between groups also reflect the fact that people with full-time jobs or self-employed are often rated as having higher creditworthiness, have better repayment ability, and therefore are granted larger loan limits. This chart shows a relatively positive picture: businesses are distributing credit in a diversified manner and covering many different customer groups. However, to optimize efficiency, lending organizations can consider strengthening support policies for the Part-time and Self-employed groups, which are potential groups but are easily limited due to unstable income factors.

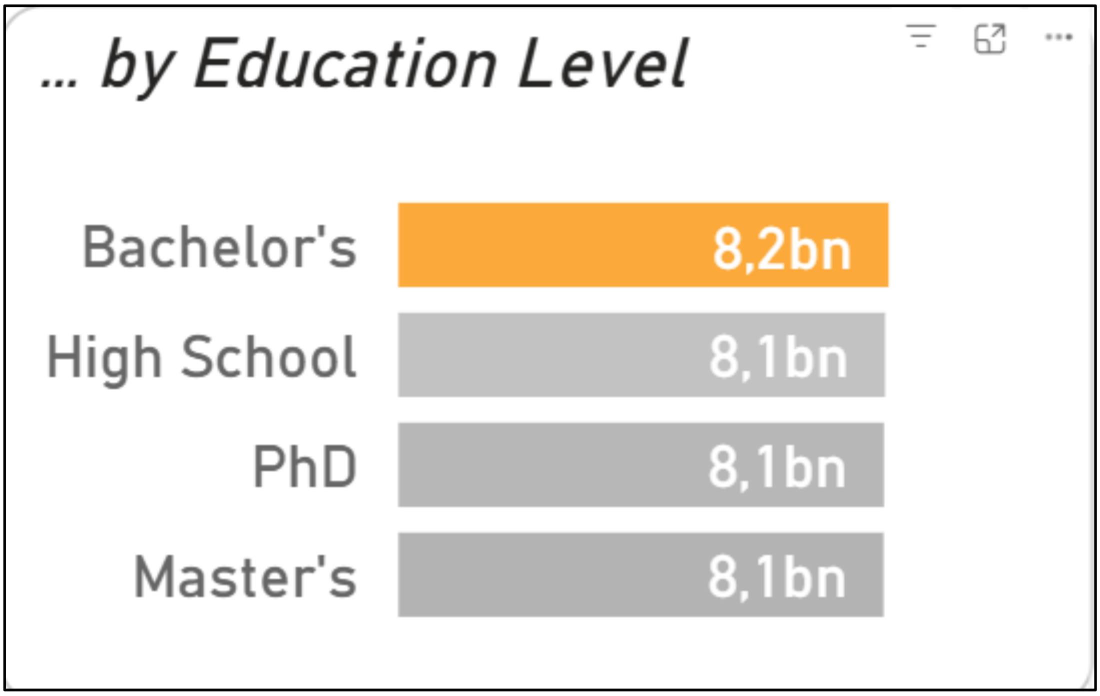

Figure 5. 16. Loan Amount by Education Level.

The Loan Amount by Education Level chart is also presented as a horizontal bar chart, in which the vertical axis represents the educational levels such as High School, Bachelor's, Master's, and PhD, and the horizontal axis represents the total loan value accordingly. The results show that the group of customers with Bachelor's degrees accounts for the highest proportion with a total loan value of about 8.2 billion VND, followed by the High School and Master's groups, while the PhD group accounts for the smallest proportion. This structure partly reflects the actual size of the labor market: the Bachelor's group accounts for the largest proportion of the labor force and is also the group with a stable income, suitable for personal credit products. In contrast, the group with postgraduate degrees, although having higher income, often has less demand for consumer loans, leading to a lower overall loan value. This chart emphasizes the role of education as an indicator of financial risk. Customers with higher education levels are more likely to maintain employment and have lower default rates, making them more likely to be granted credit with higher limits. However, businesses should also note that the High School group, despite having lower education levels, still accounts for the second largest proportion, indicating that this is a group with high actual demand for loans. Combining flexible credit policies with modern risk assessment tools will help to expand the lending target without significantly increasing the risk.

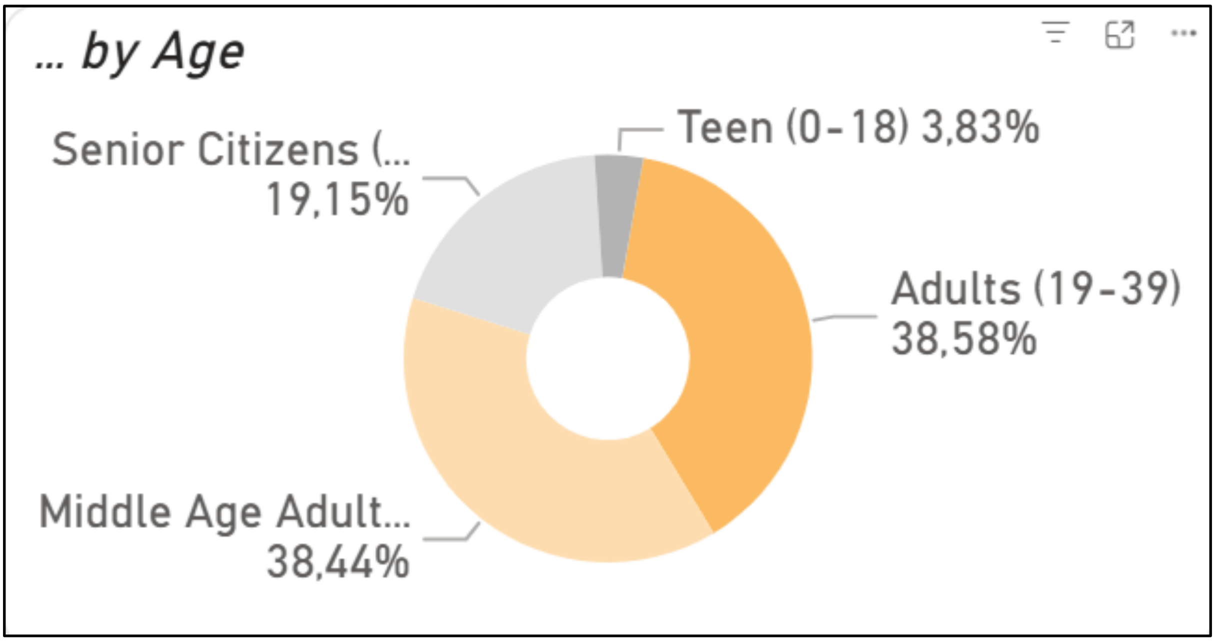

Figure 5. 17. Loan Amount by Age char

The Loan Amount by Age chart is presented in a donut chart, illustrating the proportion of loan value by four main age groups: Teen (0–18), Adults (19–39), Middle Age Adults (40–59), and Senior Citizens (59+). The results clearly show that the two groups Adults and Middle Age Adults account for almost equal proportions, 38.48% and 38.48% of the total loan portfolio, respectively, while the Senior Citizens group accounts for 19.18% and the Teen group only 3.86%. The strong focus on the two middle-aged and mature age groups shows that the business's loan portfolio is aimed at customers of active working age, with stable income and high credit demand. It can be concluded that the business is maintaining a reasonable and safe credit population structure, focusing on the group of customers with the best debt repayment ability. However, the relatively low proportion of the young group (Teen) is also a signal that the market for study credit or youth start-up credit has not been strongly exploited.

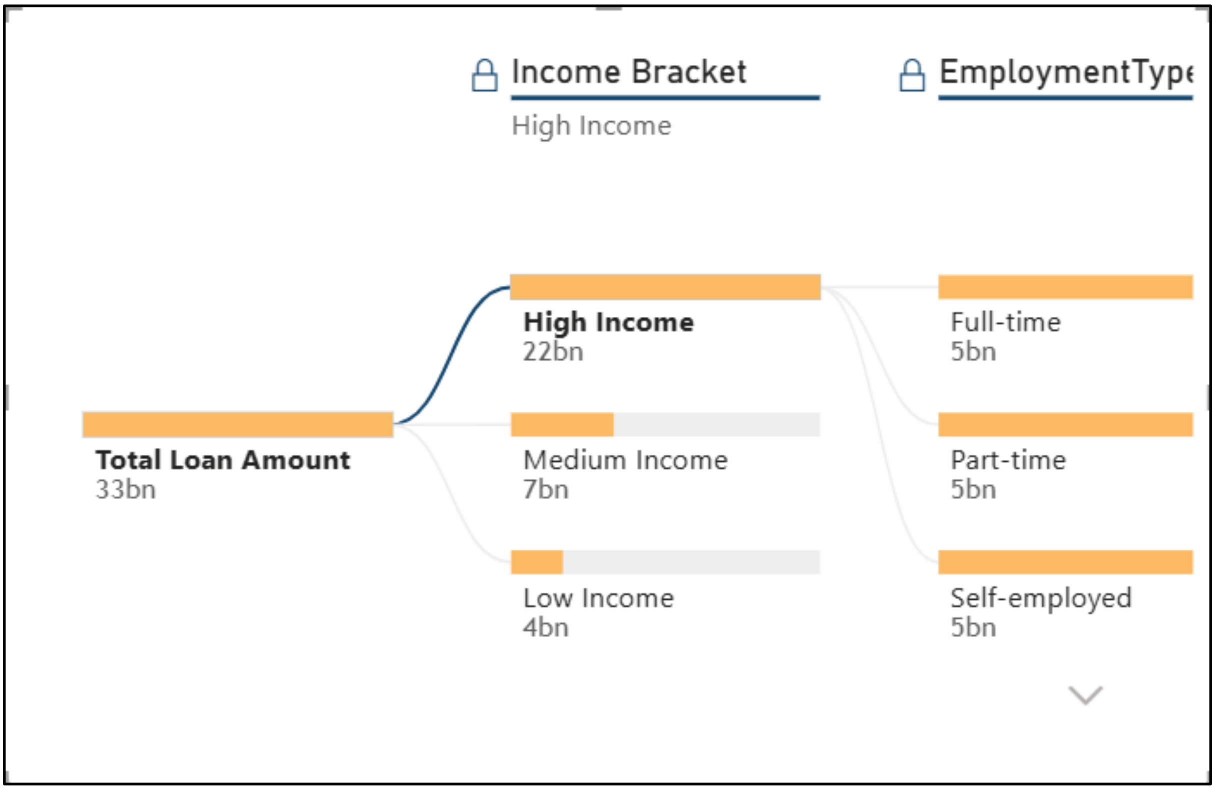

Figure 5. 18. Income Bracket & Employment Type chart

The Income Bracket & Employment Type chart is presented in the form of a Decomposition Tree, also known as a decomposition tree chart, which is an interactive visualization that allows users to drill down from the total value of the entire loan portfolio (VND 33 billion) to more detailed components. At the first level, the data is divided by Income Bracket (income groups: Low, Medium, High), then further decomposed to the Employment Type level (job type: Full-time, Part-time, Self-employed, Unemployed). When observing the chart, we see that the majority of the loan value is concentrated in the Medium Income group, with the highest proportion in the total portfolio, showing that this is the core customer group of the business. In the middle income group, individuals with full-time jobs contribute the majority of the loan value, reflecting good access to credit and high financial stability. In contrast, in the Low Income group, the loan rate mainly comes from the Self-employed group, which suggests that freelancers, despite their low income, have a higher demand for loans to serve production and small businesses. In the High Income group, the overall loan value is lower than the average group, possibly because they are less dependent on external credit sources or tend to borrow for other investment purposes.

The special feature of the Decomposition Tree chart lies in its ability to display the hierarchical relationship between factors. Through drill-down, viewers can see a clear difference in the structure of borrowing behavior: the average income group maintains a stable loan level in all types of work, while the Low and High Income groups have a significant difference between Full-time and Self-employed. The flexibility in the chart

also allows users to quickly identify customer segments with outstanding loan values or high potential risks.
The large proportion of the middle-income group is a positive sign, as this is often the safest customer group in the portfolio: having a real need for loans but still ensuring the ability to repay. However, businesses should also pay attention to potential segments such as the low-income Self-employed group, which can become a significant source of credit expansion if supported by flexible loan packages or credit guarantee policies. Conversely, it is also necessary to closely monitor loans from the high-income group, because although the loan size is smaller, this is the group that has a large impact on profit margins due to higher net interest margins.

5.2.2 Recommendations

First, businesses should restructure their interest rate and lending limit strategies to strike a balance between growth and credit safety. The trend chart results show that average interest rates remain stable, but there are signs of an adverse impact on loan values in some periods. This suggests that customers, especially in the middle-income group, are very sensitive to the cost of capital. Therefore, instead of adjusting interest rates across the board, businesses can implement a differential interest rate policy, which applies flexible interest rates based on risk profiles, credit scores, or loan product types. This approach both ensures competitiveness and encourages groups of customers with good credit histories to expand their loan size, thereby improving net profit performance.
Second, it is necessary to expand the product portfolio and potential customer base, instead of focusing only on the middle-income group and full-time workers. The Decomposition Tree chart shows that these two groups currently account for an overwhelming proportion, but this means that portfolio risk is too dependent on a specific economic segment. Businesses can expand to the Self-employed and Part-time groups through more flexible loan packages, such as cash-flow-based lending or service contract loans instead of requiring a fixed payroll. At the same time, developing more products aimed at young adults (18–25 years old) such as student loans, startup loans, or credit cards with low limits is also a way to expand the customer base while still controlling risks through small loan sizes.
Third, businesses should apply advanced data analytics to monitor the structure of the loan portfolio over time, instead of relying solely on aggregate indicators. Combining Power BI with risk prediction models or machine learning credit scoring tools will allow early identification of unusual fluctuations in borrowing behavior. Specifically, when indicators such as Average Loan Amount increase rapidly while the average Interest Rate or Credit Score decreases, it can be a warning sign of potential risks. By setting up a dynamic risk monitoring dashboard, the finance department can promptly adjust policies on limits, interest rates, or security requirements before risks turn into bad debts.

5.3 Default Rate

5.3.1 Default Rate Overview

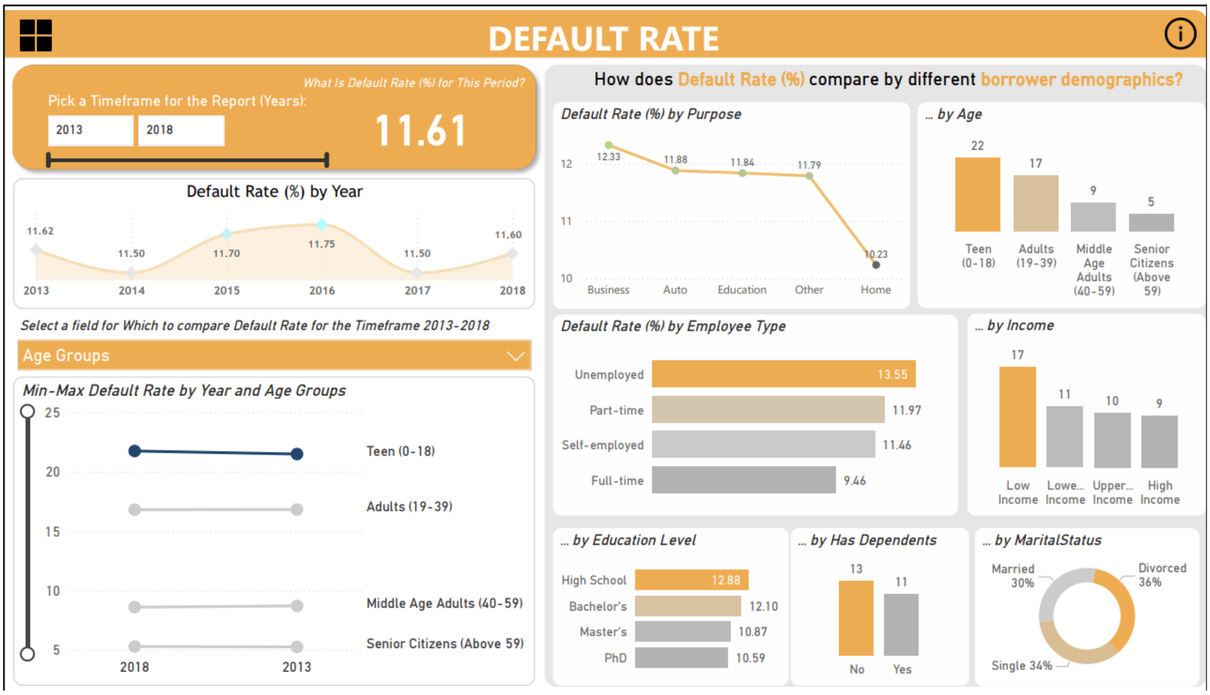

Figure 5. 19. Default Rate Dashboard

Over the period 2013–2018, the overall Default Rate averaged 11.61%, showing relative stability with only minor fluctuations between 11.50% and 11.75%, before ending at 11.60% in 2018. This consistent pattern suggests that while portfolio expansion may have occurred during these years, the overall credit quality remained steady. However, when disaggregated by borrower characteristics, significant disparities emerge, indicating uneven risk exposure across demographic segments.

From a loan purpose perspective, Home Loans exhibited the lowest default rate at 10.23%, whereas Business Loans (12.33%), Auto Loans (11.88%), and Education Loans (11.84%) performed worse. This contrast implies that loans backed by collateral (such as housing assets) tend to have stronger repayment performance, while unsecured or cash flow–based loans carry higher inherent risk. The elevated default in the business segment likely stems from market volatility and SME vulnerability during economic downturns.

When segmented by age, Teen borrowers (0–18) show an abnormally high default rate of 22%, followed by Adults (19–39) at 17%, while Middle-Aged Adults (40–59) and Senior Citizens (above 59) record far lower rates (9% and 5%, respectively). This pattern reflects a clear relationship between age and financial stability: younger borrowers likely have limited or unstable income streams, weaker credit histories, and less experience managing debt. Conversely, older borrowers may demonstrate stronger repayment discipline or access to fixed income sources, resulting in lower default rates.

Employment type analysis reinforces this observation. Unemployed borrowers recorded the highest default rate at 13.55%, compared with part-time workers (11.97%), self-employed (11.46%), and full-time employees (9.46%). This gradient illustrates the direct influence of income stability on credit performance—borrowers without regular employment are substantially more vulnerable to delinquency. It also suggests that the lending institution's credit assessment models could benefit from tighter employment verification and income consistency checks.

By income tier, Low-income groups exhibit a default rate of 17%, substantially higher than upper-income (10%) and high-income (9%) groups. This aligns with expected risk dynamics, as lower-income borrowers face greater exposure to financial shocks and reduced repayment capacity. Similarly, education level shows a modest inverse correlation with default: individuals holding only a high school degree (12.88%) default more frequently than those with Master's (10.87%) or PhD (10.59%) qualifications, likely due to differences in earning potential and financial literacy.

From a social standpoint, borrowers with dependents defaulted slightly less (11%) than those without (13%), a somewhat counterintuitive result suggesting that individuals supporting families may exhibit stronger repayment commitment. Regarding marital status, single borrowers make up the largest share (34%) of defaulters, followed by married (30%) and divorced (36%) groups, indicating that lifestyle stability and shared financial responsibilities could play a role in credit discipline.

In summary, while the aggregate Default Rate appears stable, deeper diagnostic analysis reveals that risk is heavily concentrated among young, low-income, and unemployed borrowers, as well as among unsecured loan categories such as business and education lending. This points to structural weaknesses in borrower screening and income-based risk pricing. To enhance portfolio health, the institution should tighten underwriting criteria for younger and lower-income applicants, expand collateral-backed products, and integrate behavioral credit scoring models that better capture demographic risk variations.

5.3.2 Default Rate Details (Drill through)

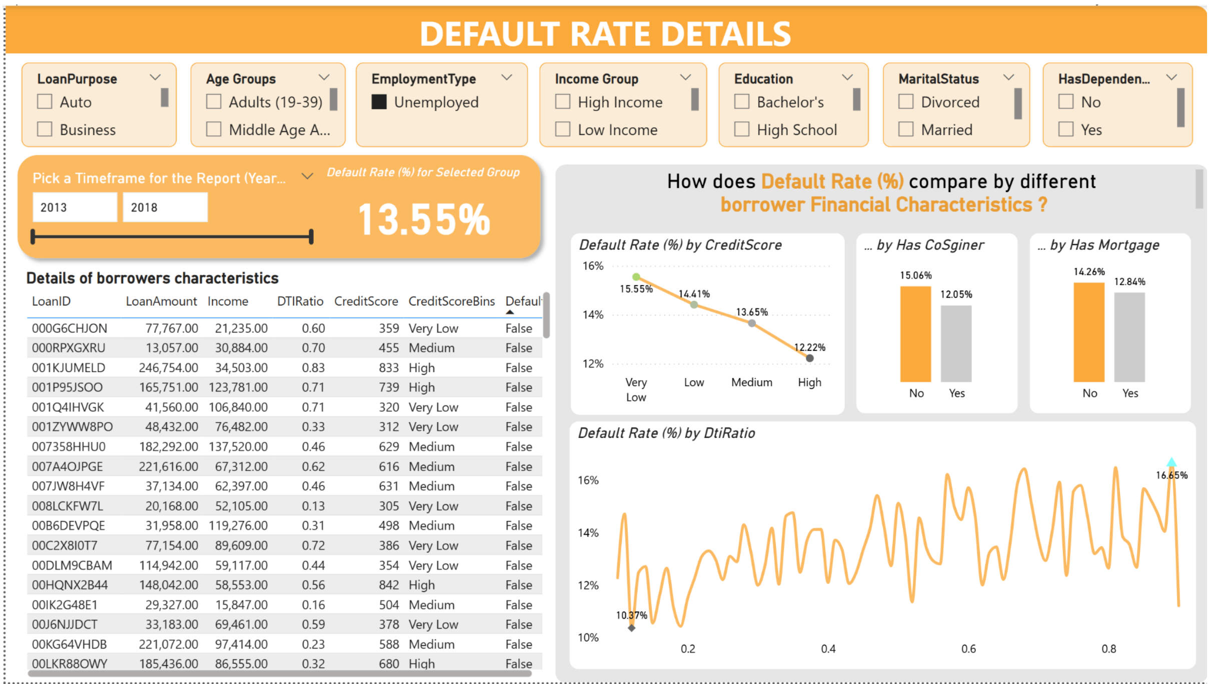

Figure 5. 20. Default Rate Details

After reviewing the overall default rate by employment type, the Unemployed group was selected for a deeper drill-through analysis since it recorded one of the highest default rate among all groups. This indicates that employment status strongly affects repayment ability, as borrowers without a stable income source are more likely to default on their loans.

Within this group, the overall default rate is 13.55%, which is considerably higher than the average. Further breakdowns reveal clear financial risk patterns:

- Borrowers with low Credit Scores and high DTI Ratios show a significant rise in default probability.
- Those without a Co-signer (15.06%) or without a Mortgage (14.26%) tend to default more often than those with additional financial support or collateral.

This suggests that while income instability is the primary cause, credit behavior and financial responsibility also play important roles. Having a Co-signer or Mortgage helps reduce default risk because these borrowers likely have stronger financial backing or motivation to repay.

In conclusion, the unemployed segment presents the highest risk profile. To mitigate this, lenders should:

- Apply stricter DTI and Credit Score thresholds for unemployed applicants.
- Encourage Co-signed or collateral-backed loans to improve repayment reliability.
- Monitor this group closely during the loan lifecycle.

5.4 Conclusion

In conclusion, the visualization chapter successfully transformed the prepared and integrated loan datasets into a set of interactive, insightful, and user-friendly dashboards in Power BI. Through a combination of key performance indicators, trend analysis, and demographic segmentation, the visualizations effectively demonstrated loan portfolio performance, borrower behavior, and default risk dynamics across multiple dimensions. The use of interactive tools such as slicers, filters, and drill-through features enhanced analytical depth and flexibility, allowing users to explore relationships within the data and uncover hidden insights. Overall, this chapter accomplished its objective of bridging data processing and analytical interpretation, providing a strong foundation for the predictive modeling and credit risk assessment developed in the subsequent chapter.

CHAPTER 6 Machine Learning Model And Insight Generation

6.1 Data Preparation for Modeling

6.1.1 EDA

	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	EmploymentType	MaritalStatus	HasMortgage	HasDependents	LoanPurpos
0	56	85994	50587	520	80	4	15.23	36	0.44	0	0	0	1	1	
1	69	50432	124440	458	15	1	4.81	60	0.68	2	0	1	0	0	
2	46	84208	129188	451	26	3	21.17	24	0.31	2	3	0	1	1	
3	32	31713	44799	743	0	3	7.07	24	0.23	1	0	1	0	0	
4	60	20437	9139	633	8	4	6.51	48	0.73	0	3	0	0	1	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
255342	19	37979	210682	541	109	4	14.11	12	0.85	0	0	1	0	0	
255343	32	51953	189899	511	14	2	11.55	24	0.21	1	1	0	0	0	
255344	56	84820	208294	597	70	3	5.29	60	0.50	1	2	1	1	1	
255345	42	85109	60575	809	40	1	20.90	48	0.44	1	1	2	1	1	
255346	62	22418	18481	636	113	2	6.73	12	0.48	0	3	0	1	0	

255347 rows × 17 columns

Figure 6. 1. Overview of Dataset

The table below presents an overview of the Loan Default Dataset, consisting of 255,347 records and 17 attributes. Each record corresponds to a single loan transaction, detailing borrower characteristics, loan details, and repayment status.

At first glance, the dataset reflects a wide demographic and financial variety. Borrower age ranges from young adults (18 years) to near-retirement (69 years), while loan amounts extend from small short-term credits to large personal or business loans up to 249,999 units. The CreditScore values, fluctuating between 300 and 849, provide a comprehensive representation of both low- and high-credit borrowers.

Other key indicators—such as MonthsEmployed, NumCreditLines, and DTIRatio—capture employment stability and debt exposure. The Default column at the end serves as the target variable, marking repayment behavior (0 = repaid, 1 = defaulted).

This rich diversity in borrower profiles and loan structures makes the dataset suitable for developing predictive models that can generalize across multiple risk patterns.

Table 6. 1. Data Type Composition

Column Name	Data Type
LoanID	object
Age	int64
Income	int64

LoanAmount	int64
CreditScore	int64
MonthsEmployed	int64
NumCreditLines	int64
InterestRate	float64
LoanTerm	int64
DTRatio	float64
Education	object
EmploymentType	object
MaritalStatus	object
HasMortgage	object
HasDependents	object
LoanPurpose	object
HasCoSigner	object
Default	int64
Loan Date (DD/MM/YYYY)	object

An examination of the dataset schema reveals a balanced mix between numerical and categorical features.

Among the 17 columns, most quantitative metrics—like Age, Income, LoanAmount, CreditScore, MonthsEmployed, and InterestRate—are stored as int64 or float64, ensuring smooth integration with statistical and machine learning algorithms.

Meanwhile, several descriptors—such as Education, EmploymentType, MaritalStatus, and LoanPurpose—are represented as object data types, reflecting their categorical nature.

The column Loan Date (DD/MM/YYYY), also of type object, contains temporal information that enables chronological trend analysis in BI visualization.

Overall, this structure provides flexibility: numerical variables feed into predictive models, while categorical dimensions enrich Power BI dashboards through filtering and segmentation.

Table 6. 2. Data Completeness Verification

Column Name	**Missing Values**
LoanID	0
Age	0
Income	0
LoanAmount	0
CreditScore	0
MonthsEmployed	0
NumCreditLines	0
InterestRate	0
LoanTerm	0
DTRatio	0
Education	0
EmploymentType	0
MaritalStatus	0
HasMortgage	0
HasDependents	0

LoanPurpose	0
HasCoSigner	0
Default	0
Loan Date (DD/MM/YYYY)	0

Before model construction, it is essential to confirm the integrity of the dataset.

The summary table demonstrates that no missing values are present in any of the 17 fields. Every feature—from financial indicators such as LoanAmount and DTIRatio to personal descriptors like Education and MaritalStatus—is fully populated.

This exceptional completeness indicates that the dataset has undergone rigorous preprocessing or originates from a highly structured source system. It eliminates the need for imputation techniques and ensures consistent data flow throughout the ETL pipeline.

Such data reliability is particularly advantageous in Business Intelligence projects, where gaps or null entries often distort aggregate insights and model accuracy.

	Age	Income	LoanAmount	CreditScore
count	255347.000000	255347.000000	255347.000000	255347.000000
mean	43.498306	82499.304597	127578.865512	574.264346
std	14.990258	38963.013729	70840.706142	158.903867
min	18.000000	15000.000000	5000.000000	300.000000
25%	31.000000	48825.500000	66156.000000	437.000000
50%	43.000000	82466.000000	127556.000000	574.000000
75%	56.000000	116219.000000	188985.000000	712.000000
max	69.000000	149999.000000	249999.000000	849.000000

	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm
count	255347.000000	255347.000000	255347.000000	255347.000000
mean	59.541976	2.501036	13.492773	36.025894
std	34.643376	1.117018	6.636443	16.969330
min	0.000000	1.000000	2.000000	12.000000
25%	30.000000	2.000000	7.770000	24.000000
50%	60.000000	2.000000	13.460000	36.000000
75%	90.000000	3.000000	19.250000	48.000000
max	119.000000	4.000000	25.000000	60.000000

	DTIRatio	Default
count	255347.000000	255347.000000
mean	0.500212	0.116128
std	0.230917	0.320379
min	0.100000	0.000000
25%	0.300000	0.000000
50%	0.500000	0.000000
75%	0.700000	0.000000
max	0.900000	1.000000

Figure 6. 2. Descriptive Statistics Summary

A statistical breakdown of the numerical features offers insight into the overall borrower population and loan characteristics.

The average borrower age is approximately 43 years, reflecting a mid-career demographic with stable employment. Mean annual income stands around 82,499, but with significant variation (standard deviation ≈ 38,963), suggesting both modest and affluent segments are represented.

Loan characteristics are equally diverse: LoanAmount averages 127,578 with extremes from 5,000 to 249,999, while InterestRate varies widely (2%–25%), highlighting different risk categories.

CreditScore averages 574 with a broad spread, confirming the presence of both prime and subprime borrowers. Meanwhile, the DTIRatio averages 0.50, showing moderate debt exposure typical in consumer lending.

Finally, the Default variable reveals that about 11.6% of borrowers failed to repay, implying a mildly imbalanced dataset—a common challenge in credit scoring tasks.

These statistical findings serve as the analytical foundation for subsequent feature selection and model training phases.

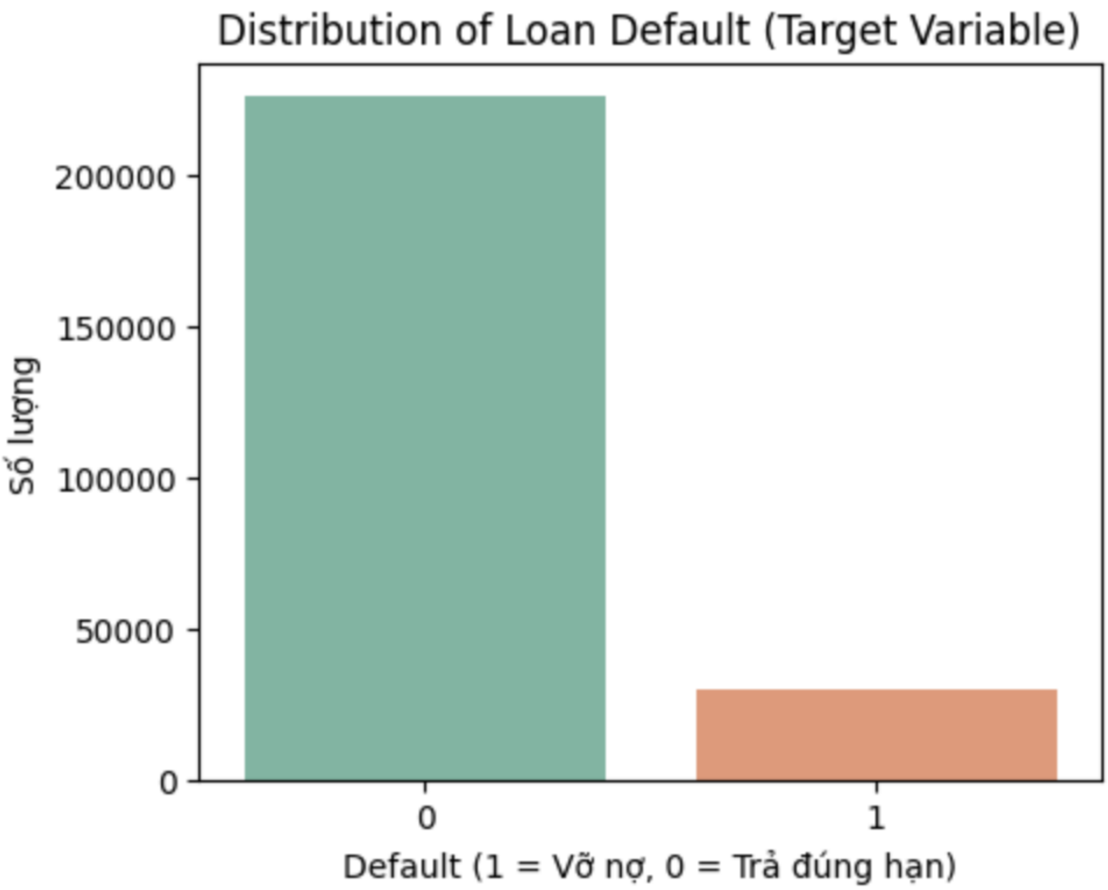

Figure 6. 3. Distribution of Loan Default (Target Variable)

The chart illustrates the frequency distribution of the **target variable – Default**, where

- 0 represents borrowers who repaid their loans on time
- 1 represents borrowers who defaulted.

The visualization shows a clear imbalance between the two classes: approximately 88.4% of the loans were repaid, while only 11.6% resulted in default.

This imbalance is typical in credit-risk datasets, as most borrowers fulfill their obligations. From a modeling perspective, however, this creates a class imbalance challenge that could bias models toward predicting the majority class ("non-default").

To ensure fair learning, techniques such as SMOTE (Synthetic Minority Oversampling Technique), undersampling, or class weighting will be applied in later stages.

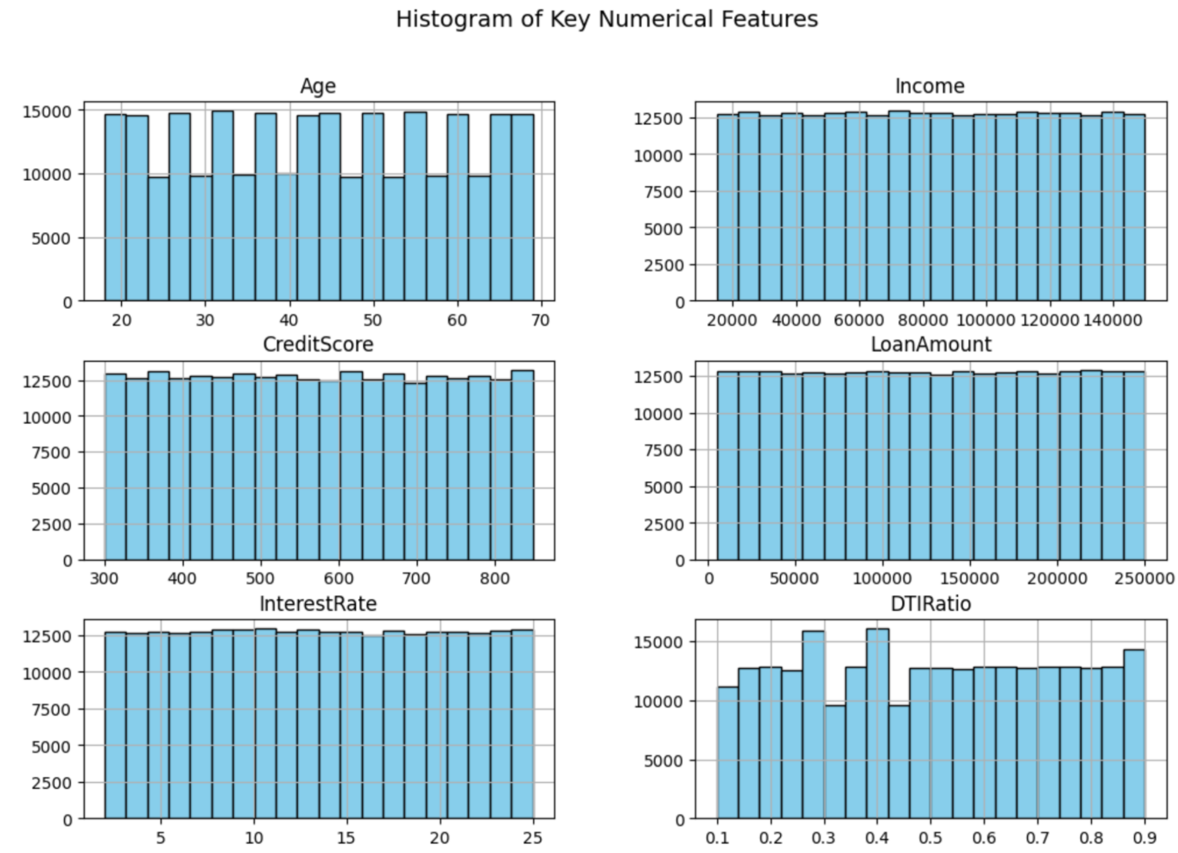

Figure 6. 4. Histogram of Key Numerical Features

A series of histograms presents the distribution of six core numerical variables — **Age, Income, CreditScore, LoanAmount, InterestRate, and DTIRatio** — to examine their spread and detect potential skewness or outliers.

- **Age** appears uniformly distributed between 18 and 69 years, suggesting the dataset includes borrowers across multiple age groups.
- **Income** and **LoanAmount** are evenly spread across their ranges, reflecting balanced representation across income levels and borrowing capacities.
- **CreditScore** covers nearly the entire range (300–850) with no sharp peaks, meaning there is no data concentration around specific credit bands.
- **InterestRate** fluctuates broadly from 2% to 25%, providing diversity in lending conditions and risk profiles.
- **DTIRatio** is concentrated between 0.3 and 0.7, indicating that most borrowers maintain moderate debt levels relative to income.

These histograms confirm that the dataset captures **a wide spectrum of financial conditions** without heavy skew, enhancing the robustness of subsequent predictive analysis.

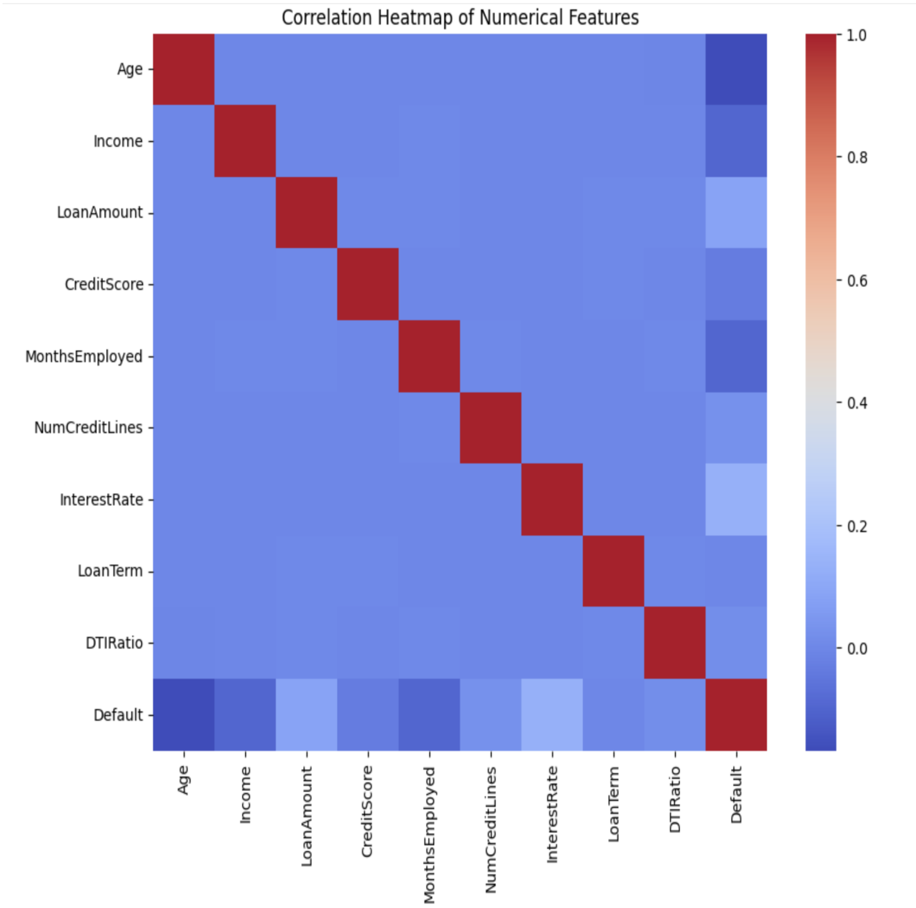

Figure 6. 5. Correlation Heatmap of Numerical Features

The heatmap visualizes the **pairwise correlation coefficients** among all numerical features, including the target variable (*Default*). Diagonal values equal to **1.0** represent self-correlation, while off-diagonal colors indicate the strength and direction of relationships. Most of the matrix appears in **cool tones (blue)**, implying **low to moderate correlations** between most features.

Some noteworthy observations include:

- **LoanAmount** and **Income** exhibit a mild positive correlation, as higher incomes generally support larger loan approvals.
- **InterestRate** shows weak correlation with **DTIRatio**, suggesting that interest rate variations are not strictly tied to borrowers' debt burden ratios.
- The variable **Default** (last row and column) has weak negative correlation with **Age** and **Income**, and a slightly positive one with **InterestRate**, indicating that younger

or lower-income borrowers and those with higher interest rates are more prone to default.

Overall, the heatmap confirms **no severe multicollinearity**, making all features suitable for inclusion in the modeling stage.

Table 6. 3. Correlation Table

Variable	**Correlation with Target (Default)**
Default	1
InterestRate	0,131273
LoanAmount	0,086659
NumCreditLines	0,02833
DTRatio	0,019236
LoanTerm	0,000545
CreditScore	-0,034166
MonthsEmployed	-0,097374
Income	-0,099119
Age	-0,167783

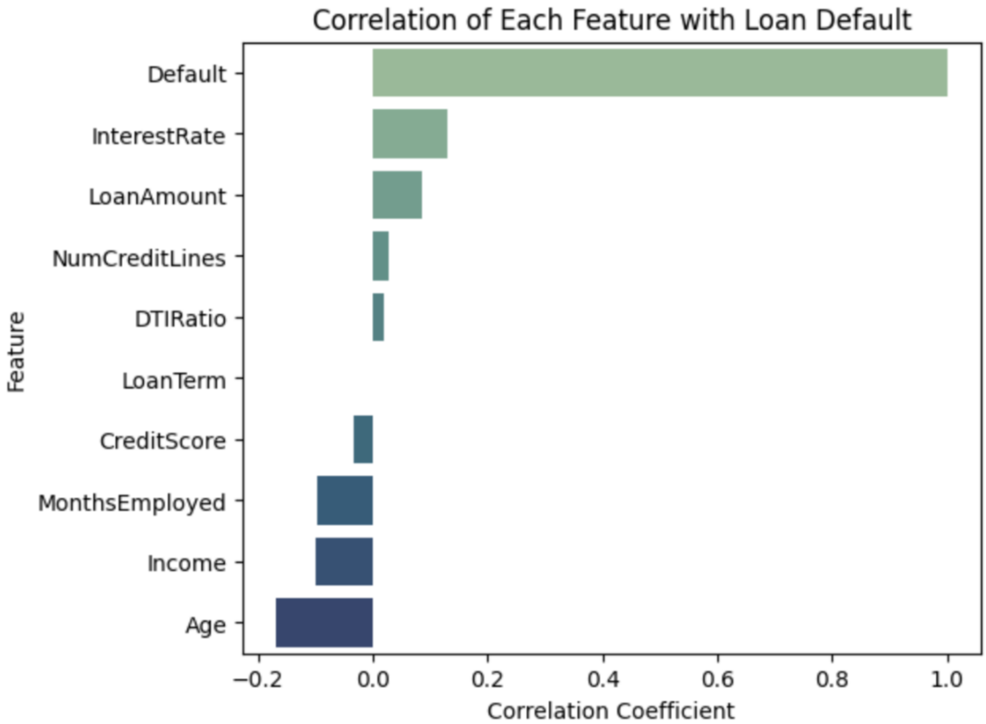

Figure 6. 6. Correlation with Target Variable

From the table and bar chart:

- **InterestRate** shows the strongest positive correlation, suggesting that higher interest rates tend to coincide with increased default risk.
- **LoanAmount** also correlates positively, implying that larger loans may be harder to repay.
- Conversely, **Age**, **Income**, and **Employment Duration** have negative correlations, meaning older, higher-income, and more stable employees are less likely to default.

Although individual correlations are not strong (|r| < 0.2), collectively they indicate meaningful trends. This weak linear relationship pattern also validates the decision to use **non-linear models (Random Forest, Gradient Boosting)** later on, as they can capture complex interactions among features better than simple linear regression.

6.1.2 Data Preparation

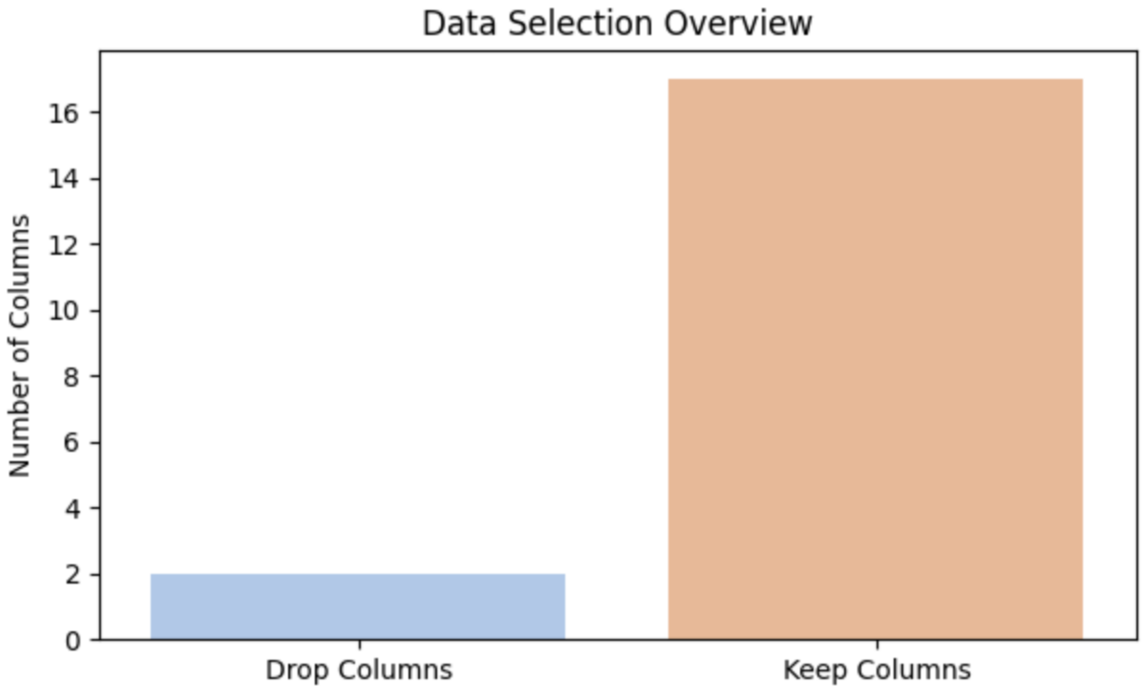

Figure 6. 7. Data Selection Overview

The bar chart provides an overview of the feature selection process carried out prior to model training.

Out of the total 17 columns in the dataset, only 2 columns were removed — LoanID and Loan Date (DD/MM/YYYY) — as these variables function solely as identifiers or timestamps and do not contribute predictive information to the model.

The remaining 15 columns were retained, encompassing both numerical and categorical attributes that capture borrower characteristics (such as Age, Education, EmploymentType, Income) and financial parameters (LoanAmount, InterestRate, DTIRatio, etc.).

This selection step ensures that only relevant features influencing loan repayment behavior are preserved, thereby improving model interpretability and computational efficiency during the training phase.

Table 6. 4. Kept and Dropped Columns

Kept Columns	**Dropped Columns**
Age	LoanID

Income	Loan Date (DD/MM/YYYY)
LoanAmount	
CreditScore	
MonthsEmployed	
NumCreditLines	
InterestRate	
LoanTerm	
DTRatio	
Education	
EmploymentType	
MaritalStatus	
HasMortgage	
HasDependents	
LoanPurpose	
HasCoSigner	
Default	

The table lists in detail the variables that were retained and excluded from the dataset after feature selection.

The kept columns include 15 features such as Age, Income, CreditScore, LoanAmount, InterestRate, and Default (the target variable).

These features collectively represent demographic information, credit standing, and loan structure, which together form the basis for predicting loan default probability.

Meanwhile, two columns — LoanID and Loan Date (DD/MM/YYYY) — were dropped, as they serve only administrative or temporal tracking purposes.

This step reflects a structured data refinement process, retaining variables with direct analytical value while eliminating redundant ones to prevent noise and dimensionality issues in the machine learning pipeline.

Figure 6. 8 Train–Test Split (80/20 Ratio)

The final figure illustrates the data partitioning strategy used for model training and evaluation.

The dataset was divided into two subsets: 80% for training and 20% for testing, aligning with standard practices in supervised learning.

This results in approximately 204,000 samples in the training set and 51,000 samples in the test set.

The training set is used to fit and optimize the machine learning models, while the test set is reserved exclusively for performance evaluation, ensuring that results reflect genuine predictive capability rather than memorization.

This split ratio maintains sufficient data diversity for model learning and guarantees reliable performance validation, especially important in imbalanced classification tasks like loan default prediction.

6.2 Model Selection and Training (Default Prediction)

6.2.1 Logistic Regression

Table 6. 5. Model Performance Summary - Logistic Regression

Metric	Value
Accuracy	0.8859
Precision	0.6203
Recall	0.031
F1 Score	0.0591

Table 6. 6. Classification Report - Logistic Regression

Class	Precision	Recall	F1-Score	Support
0 (Non-Default)	0.89	1	0.94	45170
1 (Default)	0.62	0.03	0.06	5900
Accuracy			**0.89**	51070
Macro Average	0.75	0.51	0.5	51070
Weighted Average	0.86	0.89	0.84	51070

The statistical summary outlines the model's overall performance on the test dataset.

The **accuracy** of 0.8859 indicates that the model correctly predicts loan repayment status in approximately 88.6% of cases. However, due to the **class imbalance**, accuracy alone can be misleading.

The **precision** (0.6203) shows that among all loans predicted as defaults, 62% were actual defaults, whereas the **recall** (0.0310) reveals that the model only identified around 3% of all real default cases. Consequently, the **F1-score** (0.0591) remains low, reflecting the model's inability to capture the minority class effectively.

The classification report emphasizes that the model performs well for non-default cases (class 0) but struggles with default detection (class 1). This outcome is typical for baseline models without rebalancing techniques and serves as a benchmark for more advanced algorithms later in the study.

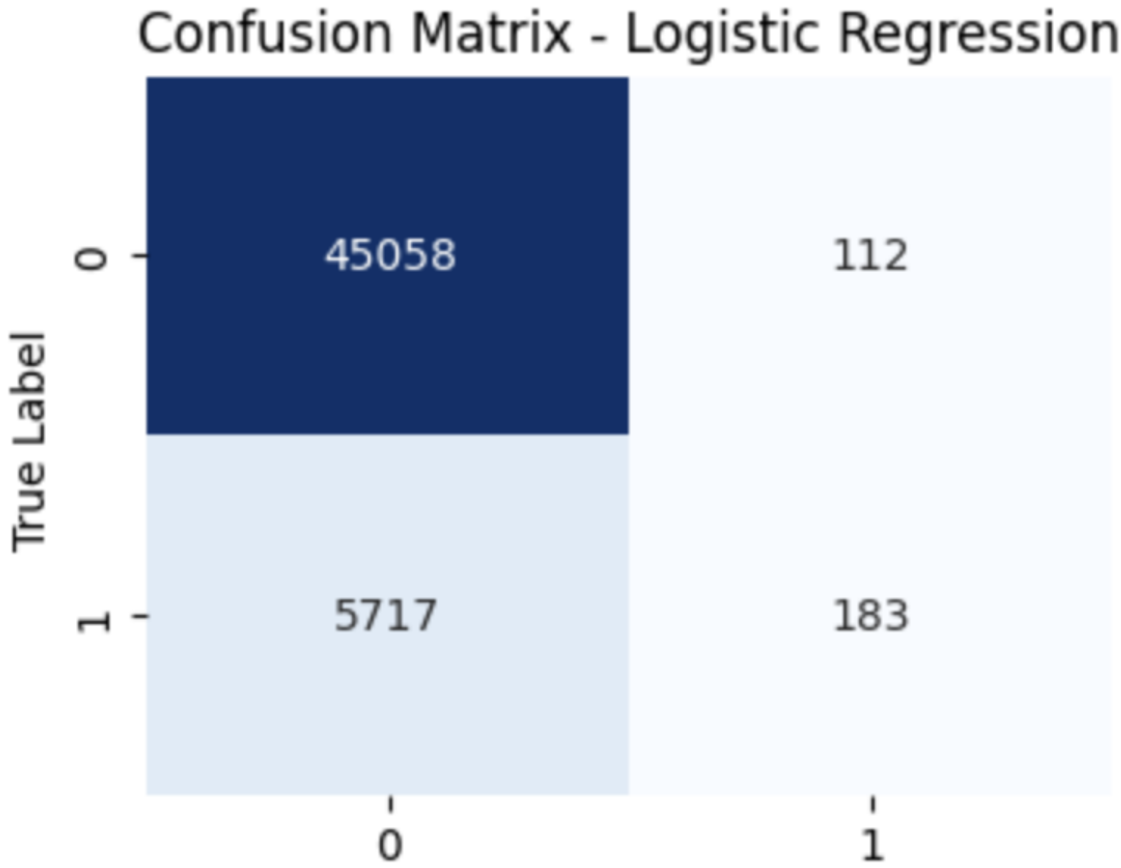

Figure 6. 9. Confusion Matrix – Logistic Regression

The confusion matrix visually summarizes the model's classification outcomes.

Out of 51,070 total test samples, **45,058 true negatives** were correctly predicted as non-defaults, and only **183 true positives** were correctly identified defaults. Meanwhile,

5,717 false negatives indicate missed default cases, and **112 false positives** represent loans incorrectly classified as defaults.

This asymmetric distribution highlights the **strong bias toward the majority class**, meaning the model tends to predict “non-default” for most cases.

While this behavior yields high overall accuracy, it limits the model’s real-world usefulness in financial risk detection, where **capturing defaults (recall)** is more critical than avoiding false alarms.

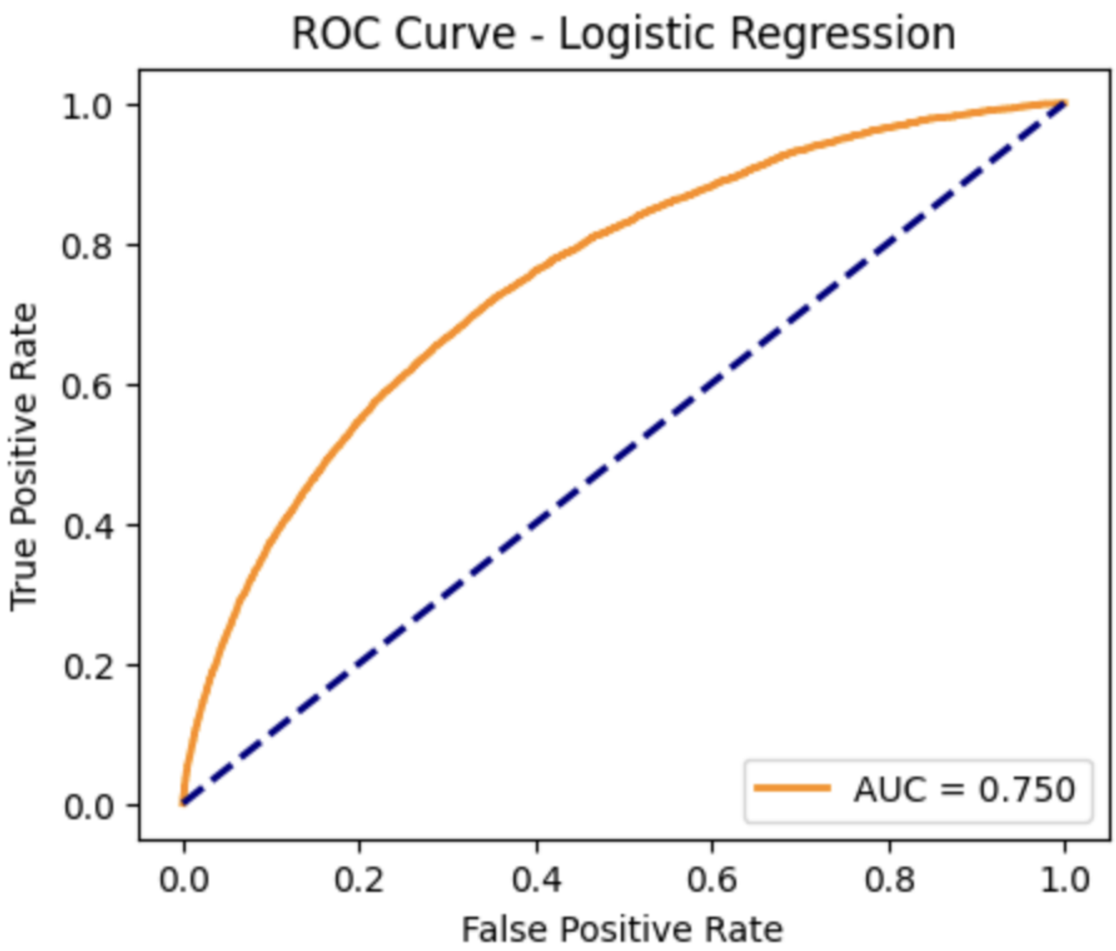

Figure 6. 10. ROC Curve – Logistic Regression

The ROC curve (Receiver Operating Characteristic) plots the True Positive Rate (Recall) against the False Positive Rate across different classification thresholds.

The AUC score of 0.750 demonstrates that Logistic Regression has moderate discriminative power in distinguishing between default and non-default borrowers.

Although not ideal for production-level credit scoring, this performance suggests the model learns some meaningful patterns from features like InterestRate, LoanAmount, and Income.

However, the curve's distance from the top-left corner confirms that the model struggles with sensitivity toward minority cases.

Future iterations may improve this by incorporating regularization tuning, feature scaling, and resampling methods (e.g., SMOTE or class weighting).

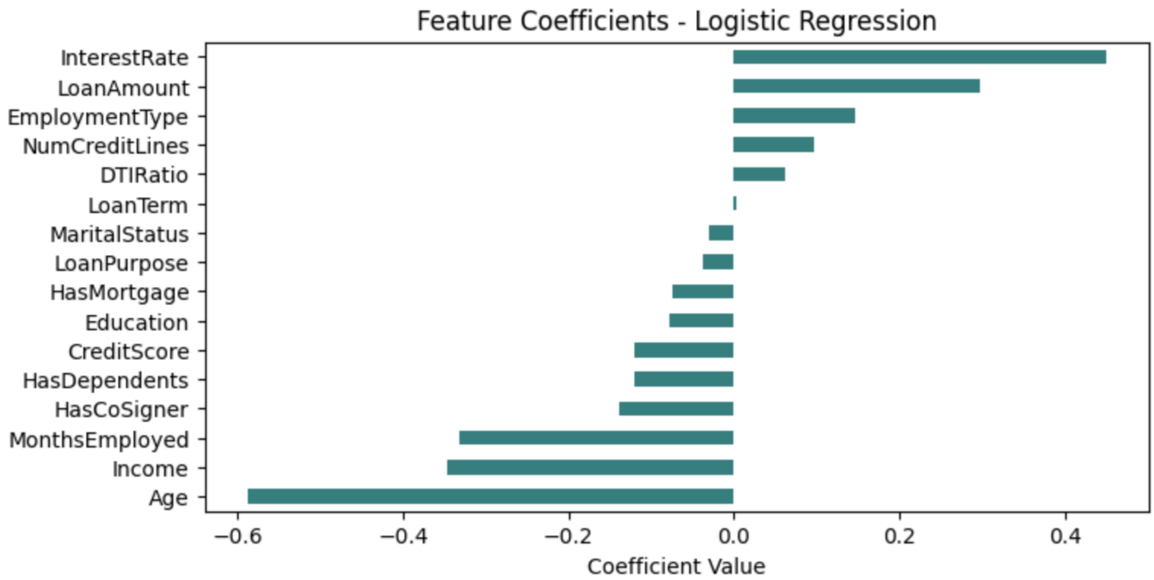

Figure 6. 11. Feature Coefficients – Logistic Regression

The bar chart visualizes the **contribution (coefficient values)** of each independent variable to the probability of default. Positive coefficients (right side) increase the likelihood of default, whereas negative coefficients reduce it. Key insights include:

- **InterestRate (+0.45)** and **LoanAmount (+0.35)** strongly increase default probability — borrowers with higher interest rates or larger loans face greater repayment pressure.
- **EmploymentType** and **NumCreditLines** also have small positive impacts, indicating that job instability and multiple credit lines may elevate risk.
- Conversely, **Age (−0.60)**, **Income (−0.35)**, and **MonthsEmployed (−0.30)** are significant negative predictors, meaning older, wealthier, and more experienced borrowers are less likely to default.

Overall, the coefficient patterns align with financial intuition, confirming that the model captures fundamental economic relationships even if predictive recall remains low.

6.2.2 Random Forest

Table 6. 7. Model Performance Summary – Random Forest

Metric	Value
Accuracy	0.8863
Precision	0.7407
Recall	0.0237
F1 Score	0.046

Table 6. 8. Classification Report – Random Forest

Class	Precision	Recall	F1-Score	Support
0 (Non-Default)	0.89	1	0.94	45170
1 (Default)	0.74	0.02	0.05	5900
Accuracy			**0.89**	51070
Macro Average	0.81	0.51	0.49	51070
Weighted Average	0.87	0.89	0.84	51070

The Random Forest model yields an overall **accuracy of 88.63%**, slightly higher than Logistic Regression, suggesting its ensemble nature captures more complex relationships between features.

Despite this, the **recall value remains low (0.0237)**, indicating limited success in identifying actual defaulters within the dataset. Meanwhile, the **precision (0.7407)** demonstrates that when the model predicts a default, it is correct about 74% of the time.

However, the **F1-score (0.0460)** confirms that the imbalance between precision and recall results in weak performance for minority detection.

The classification report reiterates this imbalance: the model is nearly perfect for non-default predictions but struggles to detect defaulters (class 1), achieving an almost negligible recall.

While Random Forest improves slightly over Logistic Regression, it still emphasizes the challenge of **class imbalance**, motivating future experiments with oversampling or tuned cost functions.

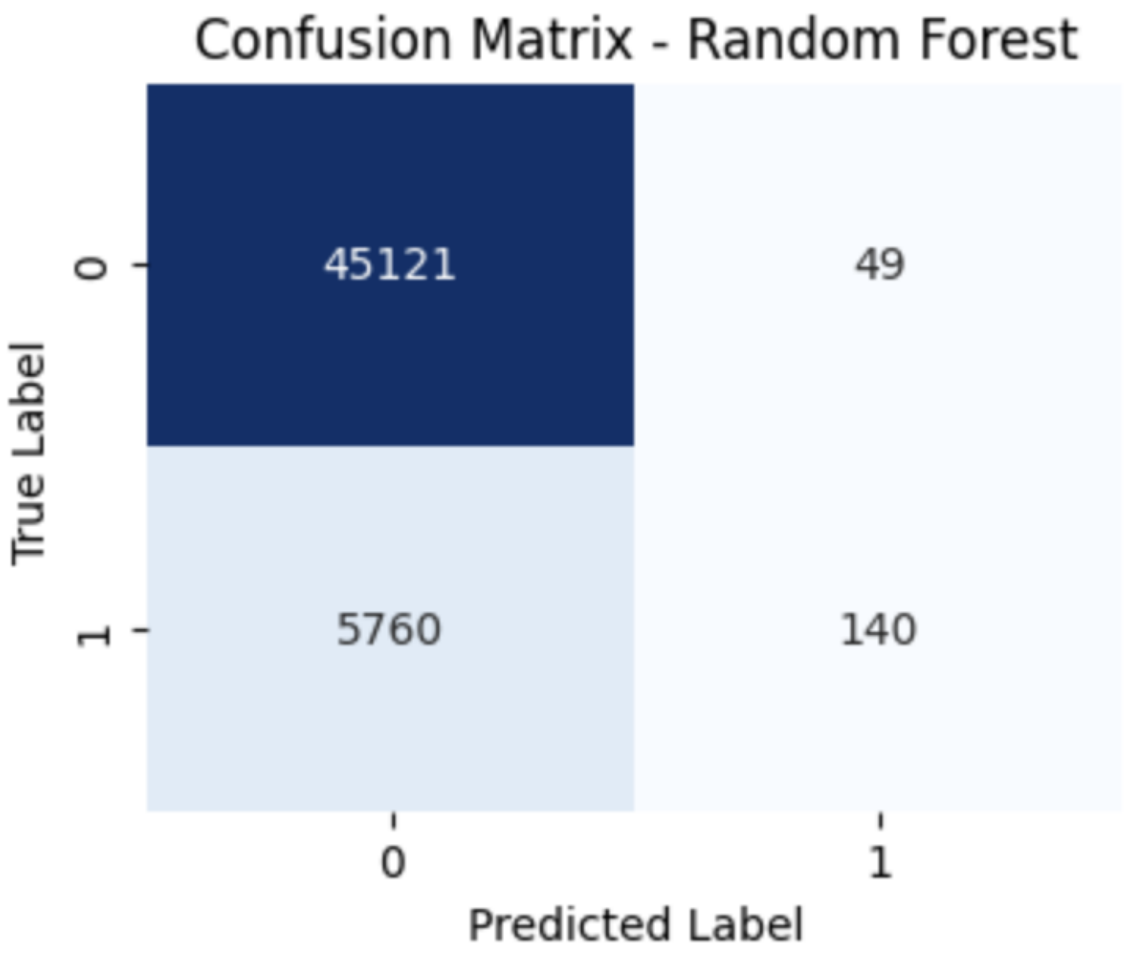

Figure 6. 12. Confusion Matrix – Random Forest

The confusion matrix offers a clear depiction of how the Random Forest model distributes its predictions.

Of 51,070 total test samples, the model correctly identifies **45,121 non-defaulters** and only **140 defaulters**, while misclassifying **5,760 actual defaults** as non-defaults. The matrix reveals that the model tends to **over-predict the majority class (0)**, a common issue when training on imbalanced data.

Nevertheless, compared with the Logistic Regression model, Random Forest exhibits a **slightly higher precision for the minority class**, showing that the ensemble approach helps reduce false positives. Even so, a significant number of false negatives remain, which limits its usefulness for **risk assessment systems** that prioritize recall (catching true defaults).

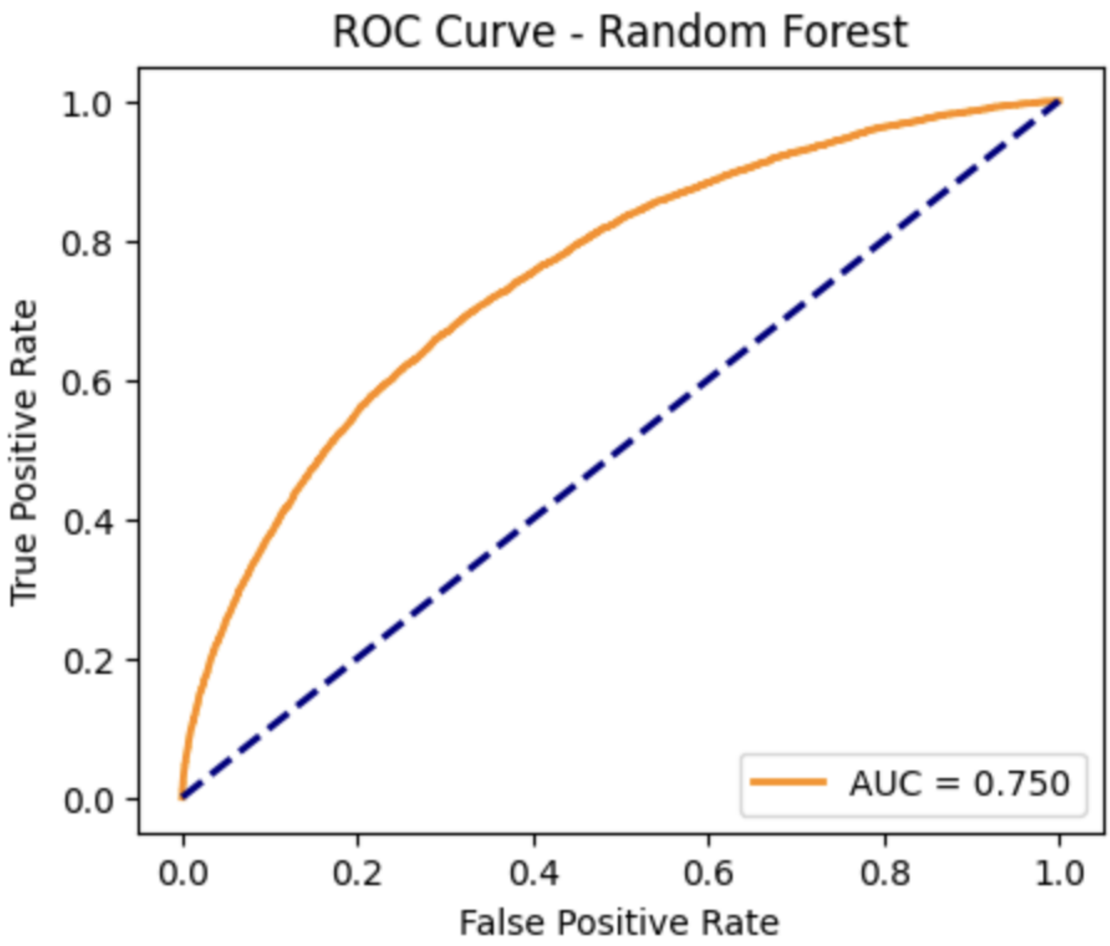

Figure 6. 13. ROC Curve – Random Forest

The ROC curve provides a graphical evaluation of the Random Forest's discriminative performance.

With an **AUC value of 0.750**, the model demonstrates **moderate separation capability** between default and non-default cases—identical in score to the Logistic Regression baseline, but achieved through a different mechanism: averaging multiple decision trees.

The curve's smooth rise toward the upper-left corner signifies that the model can effectively trade off between sensitivity and specificity depending on the threshold chosen.

This moderate AUC implies that the Random Forest model learns **useful but not decisive patterns** in borrower behavior.

Further improvement could be obtained through **feature rescaling, hyperparameter tuning (e.g., number of estimators, max depth)**, or **boosted ensemble methods** that address residual misclassifications.

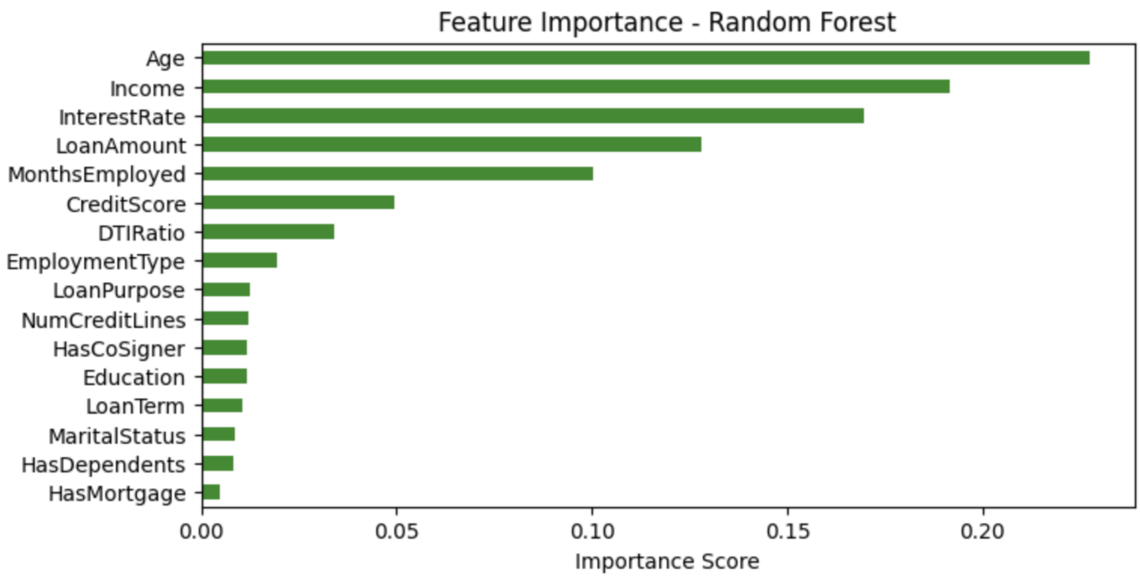

Figure 6. 14. Feature Importance – Random Forest

Random Forest provides a non-parametric measure of feature importance, showing how much each variable contributes to prediction accuracy across all trees.

The visualization highlights Age, Income, and InterestRate as the three most influential predictors of default, followed closely by LoanAmount and MonthsEmployed.

Older borrowers and those with higher income are generally less likely to default, while loans with high interest rates and large principal amounts show a stronger tendency toward default.

Variables such as CreditScore, DTIRatio, and EmploymentType also contribute meaningfully but to a lesser extent.

Other categorical features like MaritalStatus, HasDependents, and HasMortgage appear to have minor effects on prediction outcomes.

These results align with financial domain knowledge and confirm that Random Forest successfully identifies intuitive risk factors, making it a practical foundation for model refinement in later stages.

6.3 Gradient Boosting

Table 6. 9. Model Performance Summary – Gradient Boosting

Metric	Value
Accuracy	0.8873
Precision	0.6566
Recall	0.0515
F1 Score	0.0956

Table 6. 10. Classification Report – Gradient Boosting

Class	Precision	Recall	F1-Score	Support
0 (Non-Default)	0.89	1	0.94	45170
1 (Default)	0.74	0.02	0.05	5900
Accuracy			**0.89**	51070
Macro Average	0.77	0.52	0.52	51070
Weighted Average	0.86	0.89	0.84	51070

The Gradient Boosting classifier achieves an accuracy of 88.73%, the highest among the tested models, suggesting that the boosting mechanism successfully refines prediction boundaries through iterative learning.

Notably, the recall improves to 0.0515, nearly doubling that of the Random Forest, meaning the model identifies more actual default cases while maintaining a balanced precision of 0.6566.

Despite this improvement, the F1-score remains modest (0.0956), reflecting the persistent difficulty of detecting minority class samples in an imbalanced dataset.

The classification report shows excellent results for non-default borrowers (class 0) and a measurable, albeit limited, improvement for defaulters (class 1).

This demonstrates Gradient Boosting's ability to capture subtle patterns overlooked by simpler ensemble methods, making it a more promising candidate for fine-tuned financial risk prediction.

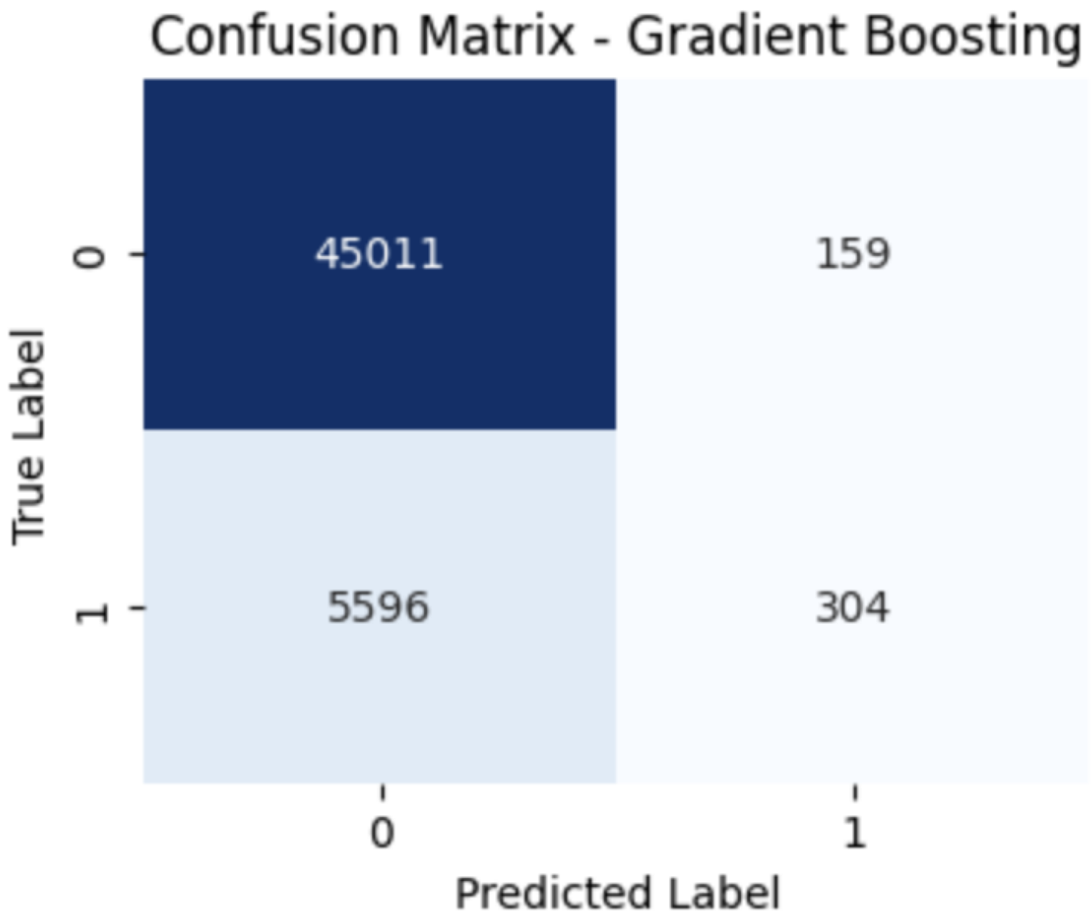

Figure 6. 15. Confusion Matrix – Gradient Boosting

The confusion matrix reveals a moderate enhancement in minority class recognition compared to previous models.

Out of 51,070 test records, 304 defaults are correctly identified, while 5,596 defaults remain misclassified, showing that Gradient Boosting succeeds in capturing more true positives than Logistic Regression and Random Forest.

The number of false positives (159) is relatively small, indicating that the model does not overcompensate by wrongly flagging non-defaulters.

This trade-off reflects Gradient Boosting's strength: it incrementally adjusts misclassified samples across multiple weak learners, leading to slightly better recall without sacrificing overall precision.

However, the imbalance remains dominant, suggesting that additional strategies such as SMOTE or threshold calibration could further enhance detection of defaulters.

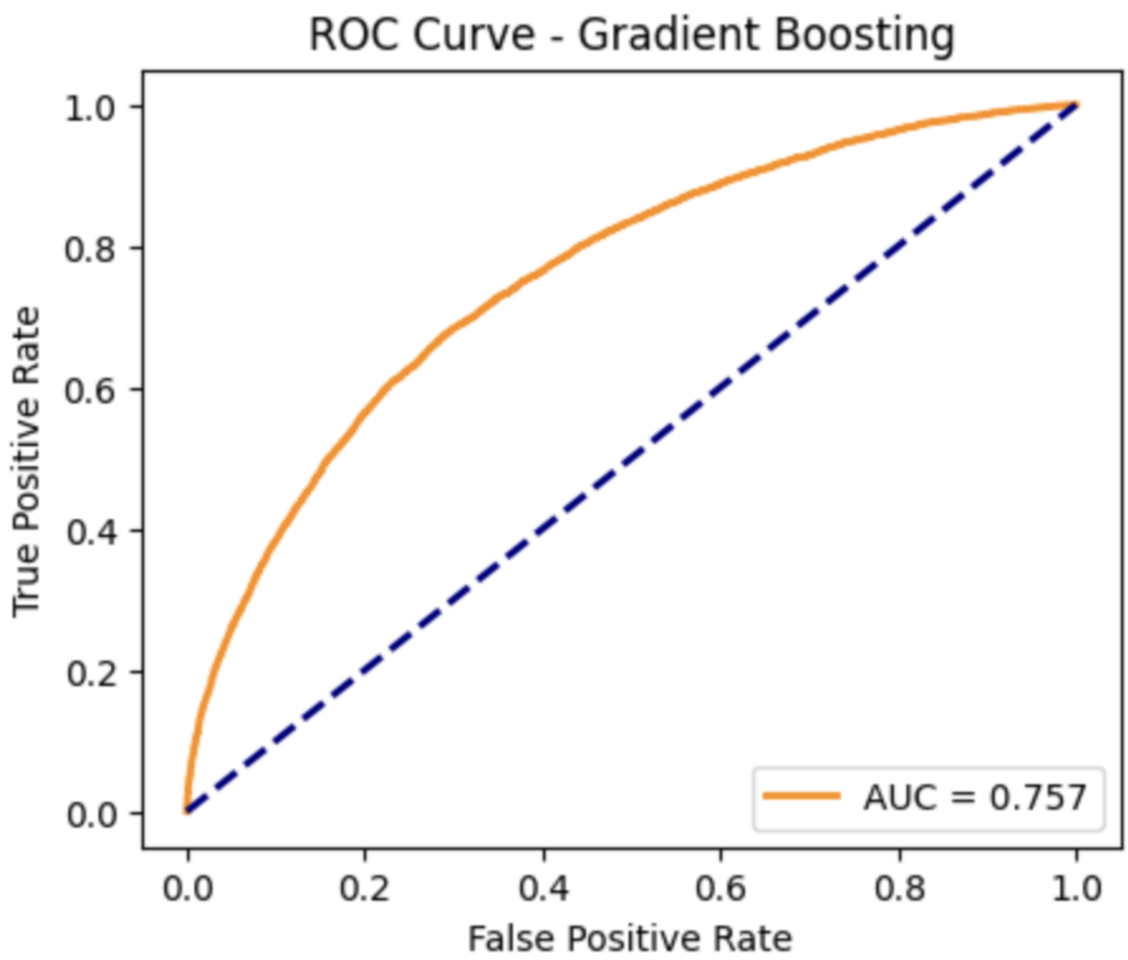

Figure 6. 16. ROC Curve – Gradient Boosting

The ROC curve for Gradient Boosting indicates a steady improvement in discriminative power, with an AUC score of 0.757.

This increase, though marginal compared to previous models, confirms that the boosting framework enhances the model's ability to rank borrowers by risk.

The curve maintains a smooth, convex shape, showing stable performance across different thresholds — a sign of effective optimization of both bias and variance.

In practical applications, this level of AUC would provide a reasonable basis for decision-making in automated credit screening systems, where prioritizing higher-risk applicants for manual review can minimize financial loss.

Overall, Gradient Boosting demonstrates the best balance between accuracy and generalization among all baseline models.

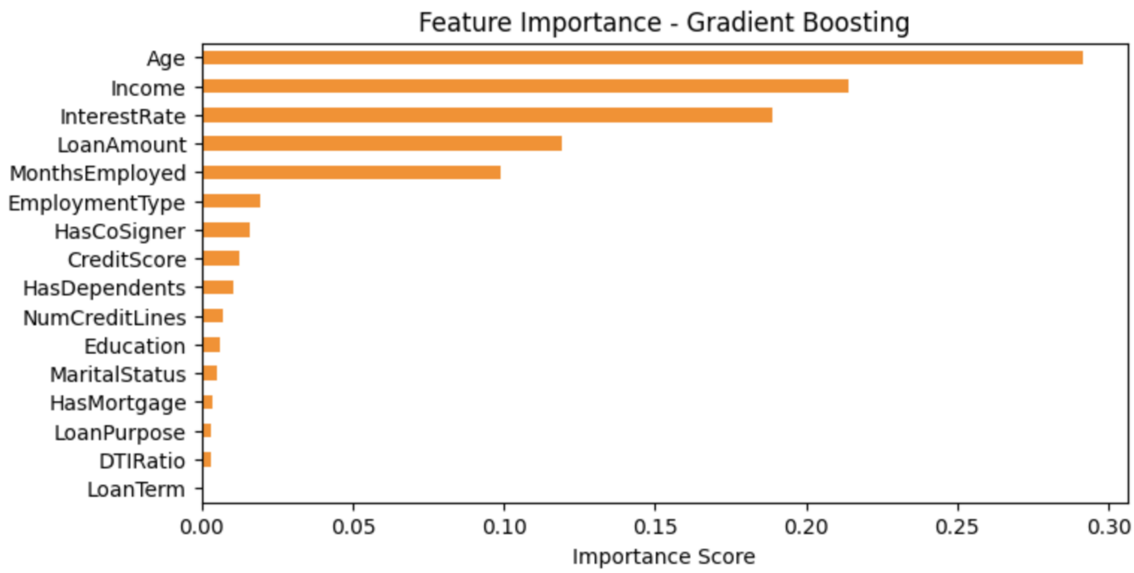

Figure 6. 17. Feature Importance – Gradient Boosting

The feature importance chart illustrates that Gradient Boosting assigns dominant influence to a few core predictors, particularly Age, Income, and InterestRate.

Age emerges as the most critical variable, followed closely by Income — indicating that younger borrowers with lower income are at significantly higher risk of default.

InterestRate and LoanAmount also play substantial roles, confirming that higher financial burden correlates with greater repayment difficulty.

Meanwhile, variables like MonthsEmployed and EmploymentType still contribute meaningfully, suggesting that employment stability remains a vital protective factor.

In contrast, attributes such as DTIRatio, HasMortgage, and MaritalStatus have minimal importance, implying that they offer less unique information after accounting for stronger financial indicators.

Overall, this distribution showcases how Gradient Boosting effectively concentrates learning on high-impact features, improving interpretability and predictive robustness simultaneously.

6.4 Model Evaluation (Accuracy, Precision, Recall, F1)

Table 6. 11. Model Performance Comparison Table

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8859	0.6203	0.0310	0.0591
Random Forest	0.8863	0.7407	0.0237	0.0460
Gradient Boosting	0.8873	0.6566	0.0515	0.0956

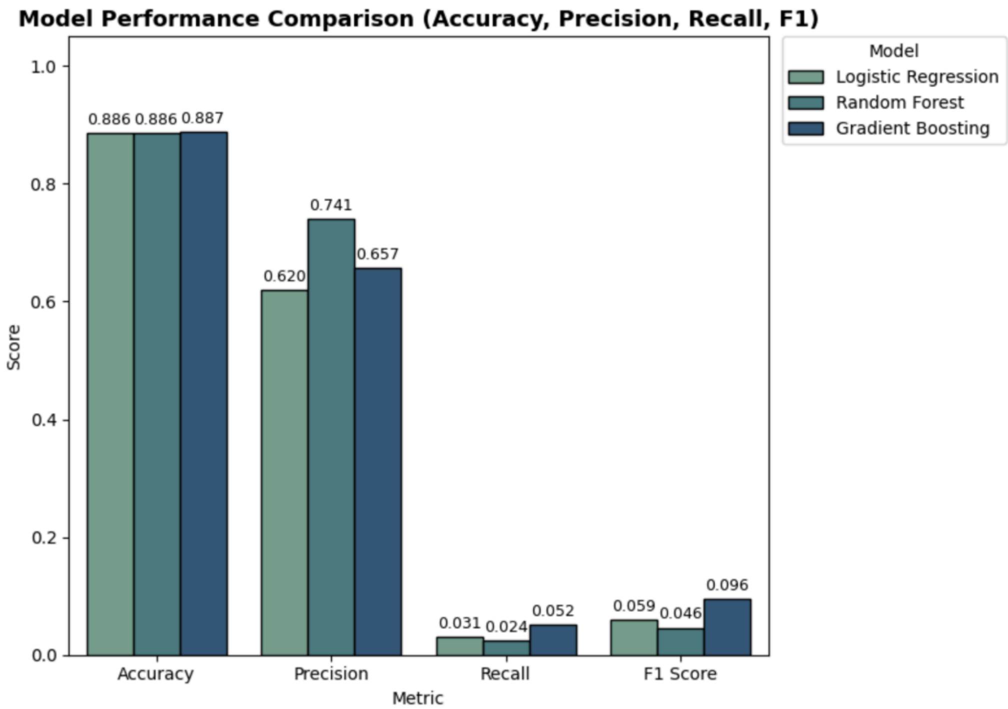

Figure 6. 18. Model Performance Comparison (Accuracy, Precision, Recall, F1)

The figure provides a detailed comparison of the three machine learning models — **Logistic Regression**, **Random Forest**, and **Gradient Boosting** — across four essential evaluation metrics: **Accuracy**, **Precision**, **Recall**, and **F1 Score**. These metrics collectively assess how well each model performs in predicting loan default cases under an imbalanced dataset setting.

The **Accuracy** scores are nearly identical for all three models, ranging from **0.8859 to 0.8873**, which suggests that each model achieves a high rate of correct predictions overall. However, in this specific problem, accuracy is not a sufficient indicator of real-world performance because the majority of borrowers do not default, leading to high accuracy even if the model fails to identify defaulters.

When comparing **Precision**, the **Random Forest model** records the highest value (**0.7407**), indicating its strength in minimizing false positive predictions — meaning that when it classifies a borrower as high-risk, it is correct about 74% of the time. This is desirable for financial institutions aiming to maintain credibility and avoid unnecessarily flagging reliable borrowers.
In contrast, **Gradient Boosting** achieves slightly lower precision (**0.6566**) but compensates with a notable improvement in **Recall (0.0515)**, which measures how effectively the model detects actual default cases. While recall remains modest due to class imbalance, this increase represents a meaningful gain over both Logistic Regression (**0.0310**) and Random Forest (**0.0237**), suggesting that Gradient Boosting captures more of the critical minority class (defaulters).

The **F1 Score**, which harmonizes Precision and Recall, provides the most balanced metric for evaluating model effectiveness in such skewed datasets. The **Gradient Boosting model** achieves the highest F1 value (**0.0956**), outperforming both Logistic Regression (**0.0591**) and Random Forest (**0.0460**). This indicates that although Gradient Boosting slightly sacrifices precision, it substantially improves recall, leading to better overall detection capability for default events.

From a practical perspective, this result implies that **Gradient Boosting offers the best trade-off between identifying true defaults and limiting false alarms**. The model's adaptive learning process, which corrects errors made by weak learners iteratively, allows it to uncover complex nonlinear relationships between borrower attributes (such as income, age, and loan amount) and repayment behavior.

In conclusion, while all models demonstrate strong general accuracy, the **Gradient Boosting classifier** emerges as the **most effective model for credit default prediction** with highest F1 Score. Its ability to identify a greater proportion of true defaulters, even under severe class imbalance, makes it particularly suitable for financial risk assessment tasks where recall and F1 are of greater operational importance than raw accuracy.

CHAPTER 7 Conclusion, Limitation and Recommendation

7.1 Conclusion

Throughout this project, we are successfully transformed complex loan data into an integrated analytical environment that supports both descriptive and diagnostic exploration. By applying advanced Power BI functionalities such as Drill Through, Decomposition Trees, dynamic slicers, and cross-filtering, the study enabled a multidimensional analysis of borrower behavior, financial attributes, and default risks. These techniques allowed the identification of key patterns—such as the strong relationship between income, employment stability, and default probability—while also revealing anomalies within specific demographic segments. Beyond Power BI, visualization in Python was employed extensively during model training, evaluation, and feature engineering, allowing for a more transparent interpretation of predictive models through tools such as feature importance plots and SHAP value visualizations. This integration not only enhanced the interpretability of machine learning results but also reinforced data-driven decision-making in credit risk analysis. In summary, the visualization component of this research bridged data engineering, analytics, and machine learning, transforming static data into actionable intelligence and demonstrating the practical power of modern BI systems in financial risk management.

7.2 Limitation

Despite achieving its analytical objectives, this study faces several limitations primarily related to data completeness and structural design. The most critical limitation lies in the absence of a Borrower ID field, which prevents tracking individual customer behavior across multiple loan records. As a result, each record in the dataset represents an independent loan instance rather than a continuous borrower profile, making it impossible to evaluate key behavioral metrics such as loan recurrence, repayment consistency, or customer lifetime risk. This constraint restricts the study to a portfolio-level rather than customer-level analysis, limiting the ability to build predictive models that incorporate borrower history or credit evolution over time.

Furthermore, several variables in the dataset are aggregated or categorical in nature, reducing granularity in important dimensions such as income distribution, loan purpose classification, and repayment behavior. The lack of transactional or time-series data, such as payment schedules, delinquency duration, or partial repayment patterns, also constrains the ability to conduct temporal modeling or default progression analysis. These limitations affect not only model precision but also the capacity to perform causal inference and longitudinal tracking of financial risk.

From a technical standpoint, the integration of Power BI Dataflow provided a scalable and maintainable data pipeline. The implementation of a refresh schedule ensured that dashboards remained synchronized with the latest data versions, improving reproducibility and operational relevance. However, because the project was developed in a controlled research environment, real-time streaming or API-based ingestion could not be implemented. As a result, the analytical outputs still reflect periodic batch updates rather than continuous data flow, which may limit responsiveness to rapidly changing financial conditions.

7.3 Recommendation

Building upon the findings and constraints identified in this study, several recommendations are proposed to enhance both the analytical framework and the quality of future research. First, it is strongly recommended that future datasets include a unique Borrower ID to enable longitudinal analysis of customer behavior. This addition would allow the tracking of repeat borrowers, loan migration patterns, and repayment consistency across multiple loan cycles, thereby enabling a transition from loan-level to borrower-level analytics. Such improvement would make possible the implementation of advanced techniques such as customer lifetime value modeling, survival analysis of default risk, and retention-based credit scoring.

Second, the data architecture should be expanded to incorporate transactional and time-series data—for example, monthly payment schedules, delinquency duration, and historical changes in income or credit score. Integrating these temporal features into the Dataflow would provide a richer foundation for predictive modeling and behavioral risk monitoring, allowing researchers and analysts to understand not only *who* defaults but *when* and *why*.

Third, on the technical side, while the current Power BI Dataflow and scheduled refresh process ensure consistency and automation, future development should move toward real-time or near real-time data integration through API or event-based streaming pipelines. This enhancement would transform the system from a retrospective reporting tool into an interactive decision-support platform, capable of responding dynamically to new credit data and market fluctuations.

Additionally, combining Power BI visualization with Python-based analytical modules proved to be an effective approach for model evaluation and feature interpretation. It is recommended to further integrate these environments through Python scripts within Power BI or Azure Machine Learning connections, allowing users to train, evaluate, and visualize model outputs within a unified interface. This hybrid approach would improve both interpretability and operational efficiency.

Finally, future research should consider expanding the analytical scope by linking internal credit data with external macroeconomic or behavioral data sources (e.g., inflation, employment indices, or digital transaction records). This would enhance the explanatory power of models and support the development of context-aware credit risk frameworks. Overall, these recommendations aim to evolve the project from a static visualization system into a dynamic, data-driven decision ecosystem that supports continuous monitoring, prediction, and optimization of lending strategies.

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